

Section 5 Gynecological Cancers

48 Cancer of the Cervix



Kirsten Jorgensen, Jose Alejandro Rauh-Hain, and Ann H. Klopp

EPIDEMIOLOGY

In the United States in 2021, the American Cancer Society estimated 14,480 new cases of invasive cervical cancer would be diagnosed, and 4,290 women would die of their disease.¹ There has overall been a decrease in age-adjusted death rates caused by cervical cancer in developed countries over the past century; however, low- and middle-income countries have not seen a significant decline and currently bear the burden of nearly 90% of global deaths annually caused by cervical cancer.² The incidence of cervical cancer has increased globally from 471,000 cases in 2000 to 529,000 in 2008 to 570,000 in 2018 and ranks as the fourth most common cause of cancer incidence and mortality worldwide.³ Additionally, although the incidence of squamous cell carcinoma (SCC) overall is decreasing, the absolute and relative incidence of adenocarcinoma is increasing, particularly in women younger than 55 years old. Reasons for this increase include updated screening technologies, screening programs, and changes in risk factors, but it is not completely understood.⁴

Cervical cancer displays significant differences in incidence and mortality around the world, starkly reflecting global inequities (Fig. 48.1). In 2018, approximately 311,000 women died from cervical cancer, with the mortality rate from cervical cancer having

the highest intercountry variation among all cancers.³ World age-standardized incidence rates (ASIRs) in 2018 ranged from 9.6 per 100,000 women in high Human Development Index (HDI) countries to 26.7 per 100,000 in low HDI countries.³ The highest incidences were found in eastern, southern, and western Africa with an ASIR of greater than 40 per 100,000, and China and India together contributed 35% of the global burden of cases and deaths. Of 185 countries assessed in 2018, cervical cancer was among the three most commonly diagnosed malignancies in women of all ages in 98 (52%) countries, and among women younger than 45 years, it was among the top three cancers in 146 (79%) countries.³ Difference in incidence and mortality rates may reflect a difference in exposure to known risk factors and inequalities in access to preventive measures, screening, and effective treatment.

The World Health Assembly adopted the Global Strategy for cervical cancer elimination in August 2020 to address global inequities in preventing and curing cervical cancer world-wide. The target for this strategy is to achieve an incidence rate of fewer than 4 per 100,000 women in all countries, and three key pillars to success will be vaccination of 90% of girls with human papillomavirus (HPV) vaccine by the age of 15 years, screening of 70% of women by age 35 years and again by age 45 years, and treatment of 90% of women with precancerous lesion and 90% of women with invasive

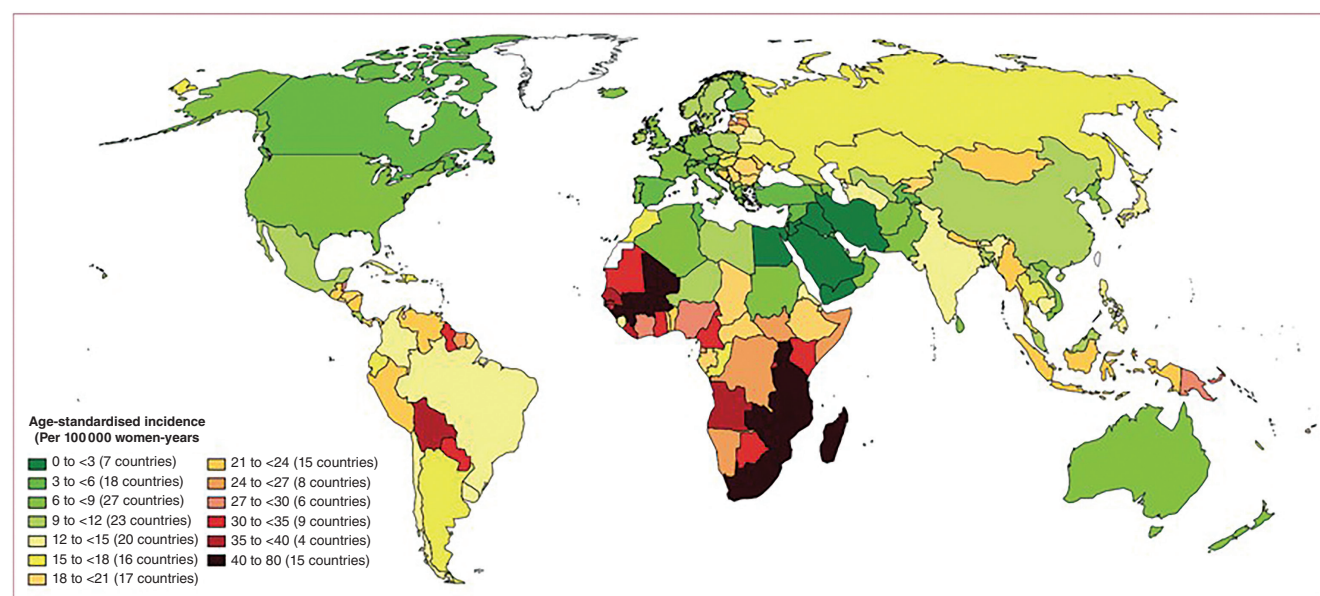


Figure 48.1 Geographical distribution of world age-standardized incidence of cervical cancer by country, estimated for 2018. (From Arbyn M, Weiderpass E, Bruni L, et al. Estimates of incidence and mortality of cervical cancer in 2018: a worldwide analysis. *Lancet Glob Health*. 2020;8(2):e191–e203. doi:10.1016/S2214-109X(19)30482-6.)

cancer. The goal is to meet the “90-70-90” targets by 2030 to eliminate cervical cancer by the end of the 21st century.²

Inequities in cervical cancer diagnosis and treatment are also seen at the country level. Despite the overall low incidence of cervical cancer in the United States, when the age-standardized rates of cervical cancer deaths were corrected for hysterectomies performed, Black women had a corrected mortality rate from cervical cancer of 10.1 per 100,000, whereas white women had a corrected mortality rate of 4.7 per 100,000.⁵ The underlying cause for racial disparities in cervical cancer is thought to be multifactorial. Prior studies have suggested differences in histology,⁵ stage at presentation,⁶ disparities in care provided by racial group,^{7,8} and cultural differences in levels of trust in the medical system may play a role. Encouragingly, the trend in mortality rate appears to demonstrate a narrowing of the gap between Black and white women, with higher decreases in mortality seen among Black women, particularly among the age groups most likely to have received the HPV vaccine.⁹

There are many theories regarding the cause in the overall trend in decreasing mortality from cervical cancer world-wide. The effectiveness of screening programs (Papanicolaou [Pap] and HPV testing) are credited with the significant decreases in incidence and mortality from cervical cancer¹⁰; however, death rates from cervical cancer proceeded routine Pap testing, suggesting that other factors may have played a role. Additionally, the advent of the HPV vaccine has demonstrated a decreased incidence in high-grade cervical dysplasia in initial studies.¹¹ It is estimated that increased vaccination rates, particularly among populations with high rates of HPV, will contribute to ongoing declines in incidence and subsequent death from cervical cancer.¹⁰

Human Papillomavirus

One of the most significant risk factors for the development of cervical intraepithelial neoplasia (CIN) and invasive carcinoma of the cervix is infection with HPV. HPV is one of the most common sexually transmitted diseases worldwide and is thought to be the most commonly transmitted disease in the United States.¹² HPV can be identified in 95% to 100% of cervical cancers, making it a necessary, though not sufficient, cause for most cases of cervical cancer.

Molecular Pathogenesis

Persistent infections with high-risk HPV genotypes (HPV 16, 18, 31, 33, and 45) have been implicated as important factors in the pathogenesis in the majority of cancers of the cervix. HPVs are non-enveloped, encapsulated, circular double-stranded DNA viruses¹³. They replicate within the basal cells of the epidermis with a relatively stable genome, thus solidifying its persistence in the epithelium. Initially, HPV infection persists in the episomal state; however, integration of the viral genome into the host genome is seen in most invasive cancers and some high-grade dysplastic lesions. Integration typically occurs in genomically fragile sites and appears to be related to underlying risk factors in the host.¹⁴

Transformation of human epithelia by high-risk HPV is thought to relate closely to the two proteins E6 and E7 that interfere with cell cycle control mechanisms, apoptosis, and chromosomal stability within the host cells. Molecular analysis demonstrates that high-risk HPV genomes that are integrated into host cell DNA enhance the progression to invasive carcinomas, possibly caused by increased expression of the oncogenes along with less regulation. HPV E6- and E7-encoding complementary DNAs (cDNAs) from the integrated oncogene appear to have much higher transforming capacity within the host cells compared with the cDNAs from episome-derived transcripts.¹⁴

At the molecular level, E6 and E7 interact directly with p53 and pRB, respectively. Recent studies demonstrate that they work cooperatively to induce significant chromosomal instability during mitosis. In addition to primary interactions with p53 and pRB, E6

and E7 oncogenes interact with other cellular proteins involved in the cell cycle and may play a role in transformation such as the p300/CBP transcriptional co-activator, the nuclear factor- κ B-responsive promoters, and inhibition of interferon- β induction. In E6- and E7- transgenic mice, after treatment to cause tumor formation, studies demonstrated E7 predominately caused tumor promotion, whereas E6 contributed more significantly during progression and malignant conversion of benign tumors.¹⁵

Human papillomavirus infection has been demonstrated to be transitory, particularly among younger women, with studies demonstrating that 70% to 90% of women younger than age 25 years clear the virus 12 months to 24 months after they are diagnosed,¹⁶ and women 45 years of age or older clear infections more quickly.¹⁷ Despite high rates of clearance of the virus, it has developed several strategies to avoid immune system identification. HPV minimizes production of antigens during quiescent phases of the virus life cycle and in early infection predominately expresses proteins that are translocated to the cell nucleus, thus voiding immune system detection. As the infection progresses, the viral capsid proteins are quickly shed from the epithelium where there is a low-density of antigen-presenting cells. These passive techniques are complemented by the activity of E6 and E7 oncoproteins that block immune-related gene expression and signaling pathways within infected keratinocytes, thereby suppressing the immune response and allowing for the promotion of cancer development.¹⁸

Clinical Pathogenesis

Human papillomavirus infection is very common, with some estimates that as many as 4 in 5 women will have had HPV at one time by the time they are 50 years old.¹⁹ The National Health and Nutrition Examination Survey from 2013 to 2016 found the prevalence of genital HPV in the among adults age 15 years to 59 years in the United States to be 38.4% among women and 41.8% among men, with an estimated 19.2 million women having a disease-associated HPV-type infection and 6.1 million women diagnosed with an incident infection of a disease-associated HPV type in 2018.²⁰ Despite a relatively high prevalence in the population, a minority of those with HPV infection go on to develop persistent infections that increase the risk of cancer development.¹⁹

In addition to the molecular mechanisms of HPV to induce cellular transformation, there are recognized clinical cofactors that may contribute to the oncogenic potential. Smoking, parity, coinfection with other sexually transmitted infections, long-term use of oral contraceptives, early initiation of sexual activity, and having multiple partners are all implicated in an increased risk of cervical cancer.^{21,22} One hypothesis for these factors' role in oncogenesis is their effect on chronic inflammation, leading to reactive oxygen species that in turn lead to host DNA damage, thus weakening host defense systems.²¹ However, many of the studies that found associations between these cofactors with either HPV infection or cervical cancer had potential significant confounding, including sexual behaviors, age of first delivery, and other risk behaviors.²² There are other studies that demonstrate protective factors for cervical cancer, including use of intrauterine devices, possibly caused by a cellular immune response triggered by these devices, and circumcision in men, leading to lower incidence of HPV and thus cervical cancer in female partners.^{23,24}

Immunodeficiency is thought to play a role in enhancing the effects of high-risk HPV. People with preexisting conditions such as human immunodeficiency virus (HIV) or iatrogenic suppression from a prior transplant are at high risk for oncogenesis if infected with HPV. Women living with HIV are more likely to contract HPV and have a sixfold higher risk of developing cervical cancer compared with women without HIV.²⁵ Cervical cancer is the most common malignancy diagnosed among women with HIV, and with the advent of antiretroviral therapy, AIDS- and HIV-related mortality rates have declined such that the life expectancy of those with HIV is nearly the same as those without, leading to a much

larger population of women at risk for development of cervical cancer over the course of their lives.²⁵

PREVENTION

Primary Prevention

The prevention of cervical cancer is best approached through multiple avenues, with key focus on primary and secondary prevention. Primary prevention, or the prevention of a disease before it occurs, is accomplished through immunization for infectious diseases. In the case of cervical cancer, the development of vaccines against the more HPV strains most likely to cause malignancy or warts is the main source of primary prevention.²⁶ The first HPV vaccine was approved by the U.S. Food and Drug Administration (FDA) in 2006 for girls and women between the ages of 9 to 26.²⁷ Since 2006, two additional HPV vaccines have been approved, the bivalent HPV vaccine Cervarix and the 9-valent HPV vaccine Gardasil-9. Since 2017, only Gardasil 9 has been distributed in the United States. All of the vaccines target HPV 16 and 18, which account for 70% of cervical cancers. Gardasil additionally protects against HPV 6 and 11, which are associated with 90% of cases of benign genital warts, and Gardasil 9 additionally covers HPV 31, 33, 45, 52, and 58 (other high-risk HPV types).²⁸ Gardasil 9 is approved for use in male and female patients ages 9 to 45, and current recommendations are for all children to be vaccinated by age 11 years to 12 years.²⁹ The Federal Advisory Committee on Immunization practices advises that those vaccinated between 9 and 14 years old receive two doses and those older than 14 years old receive three doses, a change from the original standard dose of three doses for all vaccinated individuals.^{29,30} This was reflected in recommendations by the World Health Organization (WHO) in 2014 to adopt the two-dose schedule in immunocompetent adolescents age 9 to 14 years based on noninferiority trials, cost savings, and implementation advantages.³¹ In 2018, the U.S. FDA approved the use of the Gardasil 9 to include individuals ages 27 years to 45 years.³²

The approved vaccines are all highly effective, producing antibodies to the HPV viral capsid protein and titers above those achieved with natural infection. In addition, the durability of prevention against cervical cancer for Gardasil and Cervarix appear to be at least 5 to 10 years, respectively, with ongoing studies evaluating long-term efficacy of Gardasil 9.³³ For women and girls with no history of HPV infection before vaccination, Cervarix and Gardasil appear to reduce the risk of CIN 1, 2, and 3 caused by vaccine-specific subtypes by more than 95%,³⁴ and Gardasil 9 appears to have similar effect in early studies.³³ Several observational studies suggest that even one dose of an approved vaccine may be effective against vaccine-type HPV infections, with a systematic review demonstrating particular effectiveness against HPV 16 and 18 infections. The review found one dose of an approved vaccine resulted in significantly lower HPV rate than in unvaccinated control participants, though protective antibody levels were lower in individuals who received one dose than in unvaccinated control participants with natural infection.³¹ Data are mixed regarding the use of the vaccines as therapeutic measures, with one randomized trial of 1,711 women suggesting vaccination does not appear to change the rate of clearance of the HPV virus among women who are seropositive for HPV at the time of vaccine administration³⁵ and others suggesting possible prevention of recurrent CIN 2 and 3 disease associated with HPV infection after HPV vaccination after treatment for CIN (loop electrosurgical excision procedure [LEEP]).^{36,37} Trials are underway to develop therapeutic vaccines with different mechanisms of action than currently approved vaccines, though none of these vaccines have been licensed for clinical practice.³⁷

Despite effectiveness in vaccines, there are significant barriers to their global distribution. As of 2020, only 8% of low- and middle-income countries (LMICs) have introduced HPV

vaccination schedules,³⁷ yet LMICs account for the majority of the cases of cervical cancer; thus, even 100% vaccination rates in high-income countries will not substantially decrease the global burden of cervical cancer disease.³³ Encouragingly, there has been increased number of introductions of HPV vaccination among LMICs since 2019, responding to the WHO global strategy for the elimination of cervical cancer by 2030 wherein one pillar is achieving 90% coverage target for vaccination. There is still a long way to go to achieve this milestone because only 5 countries had achieved the 90% mark, 22 achieved 75% or higher, and 25 had achieved 50% or less as of mid-2020, with a global estimate of approximately 15% of girls and 4% of boys vaccinated fully and 20% and 5% with at least one dose, respectively.³⁸ The rate of HPV vaccination lags behind vaccination rates globally for other childhood vaccines.³⁸

Secondary Prevention

Secondary prevention efforts are focused on early detection and treatment of patients with the disease, in this case, screening for HPV and cytologic abnormalities via the Pap smear. Recent studies have suggested primary HPV screening alone versus Pap cytology demonstrates greater sensitivity with some loss in specificity for detecting CIN 2+ lesions,³⁹ subject to less human error given it can be automated, and possibly more cost effective than Pap and HPV co-testing or Pap screening alone.⁴⁰

The Pap smear test for cervical cytology has been a key factor in reducing rates of cervical cancer, particularly in high-income countries with effective screening programs. Current cervical cancer screening guidelines in the United States are recommended by the American Society of Colposcopy and Cervical Pathology (ASCCP), endorsing U.S. Preventive Services Task Force guidelines established in 2018. Recommendations are to screen women ages 21 years to 65 years, with screening every 3 years with cytology alone from age 21 years to 29 years followed by cytology alone every 3 years age 30 years to 65 years or cytology and HPV co-testing every 5 years. Additionally, screening is not recommended for those older than 65 years with a history of negative Pap test results (at least the most recent three tests) or negative HPV test results (most recent two tests), provided they have no history of high-grade dysplasia (CIN 2 or 3) or cancer in the past 20 years. If a woman presents at older than 65 years old with no prior screening, the current recommendation is for co-testing of Pap and HPV. If a woman had a total hysterectomy and no history of CIN 2 or greater dysplasia, no further Pap or HPV testing is recommended.⁴¹

PATHOLOGY

Epithelial Cell Abnormalities (Pre-invasive)

Several systems have been developed for classifying premalignant cytologic and histologic cervical findings (Table 48.1). The Papanicolaou class system and CIN dysplasia systems were used before the establishment of the Bethesda system and resulted in lack of standardized reporting of clinically relevant findings.⁴² In response to this, after a 1988 National Cancer Institute Conference, the Bethesda system of classification was developed and revised in 1991.⁴³ The most current Bethesda guidelines were updated in 2014.⁴⁴ These guidelines are used in the majority of laboratories in the United States (>60% as of 2016, with another 20% planning introduction).⁴⁵ The Bethesda system classifies specimens by adequacy for evaluation (satisfactory or unsatisfactory) and then categorizes into general categories of negative for intraepithelial lesion or malignancy, epithelial cell abnormality, or other.

Further classification within the epithelial cell abnormalities includes distinguishing squamous and glandular cell abnormalities and other malignant neoplasms. The system classifies squamous

TABLE 48.1

Comparison of Cytology Classification Systems for Cervical Neoplasms

Bethesda System	Dysplasia or CIN System	Papanicolaou System
Within normal limits	Normal	Class I
Infection (specify organism)	Inflammatory atypia (organism)	Class II
Reactive and reparative changes	—	—
Squamous cell abnormalities	—	—
Atypical squamous cells of undetermined significance	Squamous atypia	Class IIR
Low-grade squamous intraepithelial lesion	HPV atypia Mild dysplasia CIN 1	
High-grade squamous intraepithelial lesion	Moderate dysplasia CIN 2 Severe dysplasia CIN 3 Carcinoma in situ	Class III
Invasive squamous carcinoma	Invasive squamous carcinoma	Class IV
		Class V

CIN, cervical intraepithelial neoplasia; HPV, human papillomavirus.

cell abnormalities into atypical squamous cells of undetermined significance (ASCUS), atypical squamous cells that cannot exclude a high-grade lesion (ASC-H), low-grade squamous intraepithelial lesion (LSIL), and high-grade squamous intraepithelial lesion (HSIL).⁴⁴ The term ASCUS was first introduced by the Bethesda

system and is now the most commonly reported abnormal Pap test result in the United States, accounting for 1.6% to 9% of Pap tests overall.⁴⁶ Using the Bethesda system, biopsy specimens can then be scored as CIN 1, 2, or 3. The term *cervical intraepithelial neoplasia* specifically refers to lesions that may progress to invasive carcinoma. The important features distinguishing CIN 1, 2, and 3 are cellular immaturity and disorganization, nuclear abnormalities, and increased mitotic activity⁴⁷ (Fig. 48.2). Lesions involving the lower third of the epithelium are generally classified as CIN 1, those in the lower and middle thirds are CIN 2, and those involving the upper third are CIN 3. Typically, LSIL corresponds to CIN 1, is often associated with low-risk HPV types, and has a low likelihood of progression to invasive cancer, whereas HSIL correspond to CIN 2, 3 and are associated with high-risk HPV.⁴⁷

Within the glandular cell abnormalities, atypical endocervical, endometrial, or glandular cells and whether neoplastic features are favored are reported.⁴⁴ Atypical glandular cells (AGCs) on screening cytology are considered high-grade lesions (>25% risk of progression to CIN 3 or invasive disease). The risk of progression to invasive disease is higher if co-testing results with HPV are positive in conjunction with AGC cytology result.⁴⁸ AGCs are characterized by morphologic changes within glandular cells beyond what would be considered typical of a benign process but are insufficient for diagnosis of adenocarcinoma in situ (AIS) and are subdivided into not otherwise specified (NOS), favoring endocervical AIS, or adenocarcinoma.⁴⁹

Adenocarcinoma in situ is a precursor of invasive cervical adenocarcinoma. Pathologic diagnosis is notable for tall, irregular columnar cells; stratified, hyperchromatic nuclei; apoptotic bodies; and increased mitotic activity with a preserved normal branching pattern of the endocervical glands and no distinct stromal invasion. It is related to high-risk HPV types (more commonly HPV 18), and 30% to 60% of women with AIS have simultaneous squamous CIN. It may be multifocal in up to 15% of patients, and endocervical AIS is the most frequent subtype. Additionally, it has several benign mimics, including reactive glandular atypia, mitotically active endocervical glands, tuboendometrioid metaplasia,

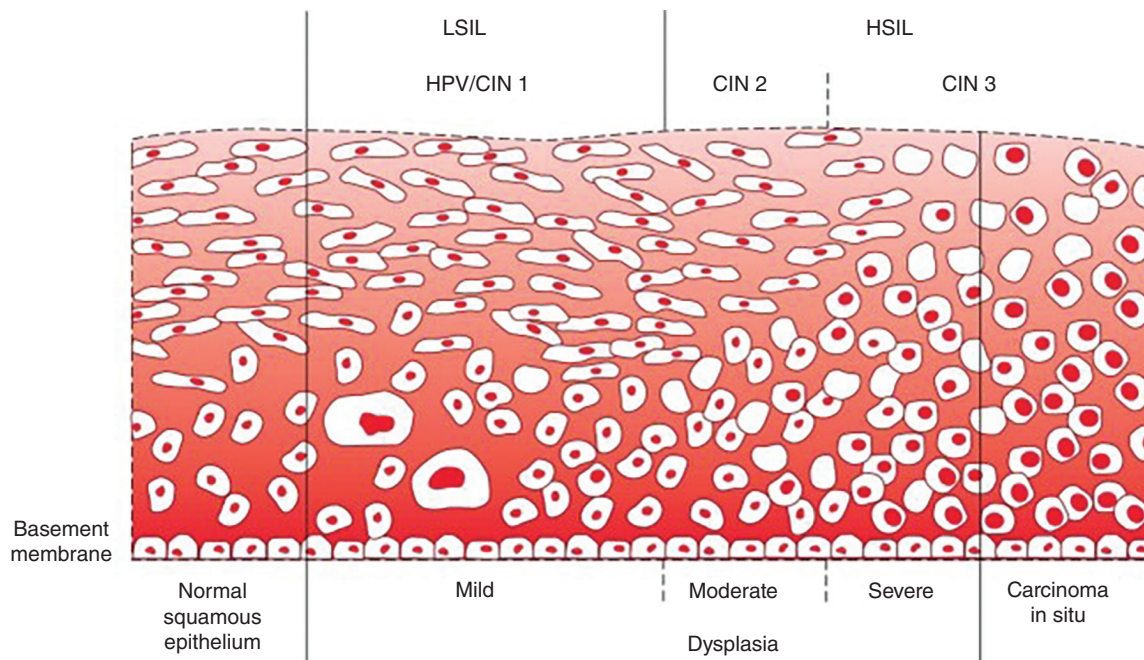


Figure 48.2 Histologic progression from normal epithelium to carcinoma. CIN, cervical intraepithelial neoplasia; HPV, human papillomavirus; HSIL, high-grade squamous intraepithelial lesion; LSIL, low-grade squamous intraepithelial lesion. (With permission from Salcedo, MP; Phoolcharoen, N; Schmeler, K. 29 - Intraepithelial neoplasia of the lower genital tract (cervix, vagina, vulva): Etiology, Screening, Diagnosis, Management. In: *Comprehensive Gynecology*. 8th ed. Elsevier; 2022:637-647.e2.)

endometriosis, mesonephric remnant intestinal metaplasia, Arias-Stella reaction, and atypical oxyphilic metaplasia; therefore, expert pathologic evaluation is advised.⁵⁰

Invasive Carcinoma

Microinvasive carcinoma is defined by the International Federation of Gynecology and Obstetrics (FIGO) as “invasive carcinoma that can be diagnosed only by microscopy, with a maximum depth of invasion of <5 mm.” This definition was updated in the 2018 revised staging (Table 48.2) to reflect that the horizontal dimension of a microinvasive lesion is no longer considered.⁵¹ After the advent of cytologic screening for early detection, the proportion of invasive carcinomas meeting microinvasive criteria increased more than 10-fold to approximately 20% in the United States.⁵² Microinvasion pathologically initially appears as blurring of the stromal epithelial junction, with protrusion of cells into the stroma. The neoplastic cells have abundant cytoplasm, hyperchromatic nuclei, and prominent nucleoli and exhibit loss of polarity at the stromal epithelial junction. Early microinvasion is usually characterized by formation of pseudoglands (nests of invasive carcinoma that can mimic crypt involvement) or scalloping or duplication of the neoplastic epithelium.⁵³ The depth of invasion is measured from the base of the epithelium (surface or gland) from which the tumor arises to the deepest point of invasion. Measuring the depth of invasion can be difficult for adenocarcinomas because they may originate anywhere within the glands, making reproducibility of measurements challenging. Current consensus recommends for adenocarcinoma the depth of invasion should be measured from the nearest surface epithelium, equating to the tumor thickness.⁵⁴

Squamous cell carcinomas account for three-fourths of cervical cancer cases in the United States, with further subdivisions into large cell keratinizing and nonkeratinizing. Nonkeratinizing carcinomas are characterized by squamous cells with hyperchromatic nuclei and moderate cytoplasm that grow within discrete nests, with potential for differentiation and degeneration at the center of the nests, whereas keratinizing carcinomas have very hyperchromatic nuclei; eosinophilic cytoplasm; and irregular, invasive nests with central “pearls” containing keratin. There are other uncommon variants of SCCs, including verrucous and papillary variants. Verrucous carcinoma has a warty appearance that macroscopically resembles condylomas with potential for extensive local invasion, though it rarely metastasizes. Microscopically, frondlike papillae can be seen with marked inflammatory reaction at the epithelial stromal junction and invasive nests of epithelium alongside well-circumscribed nests. Papillary SCCs may range from well to poorly differentiated and may be mistaken for transitional cell carcinomas of the urinary tract, requiring differentiation via immunohistochemical testing.⁵⁵

The pathologic classification of invasive adenocarcinoma has been a recent subject of discussion within the field of pathology. Historical classification by the WHO in 2014 classified based on morphologic characteristics, particularly cytoplasmic features. This schema has been questioned due to vague definitions and lack of biological basis for classification. An alternative classification system was proposed in 2018, the International Endocervical Adenocarcinoma Criteria and Classification (IECC). This classification system categorizes lesions based on underlying etiology and biologic behavior and sorts them into HPV related and HPV unrelated, with further subcategorizations of rare types. According to the IEACC, HPV-related endocervical adenocarcinomas include mucinous, usual type; villoglandular, intestinal type; and signet ring type adenocarcinomas of the WHO classification, and HPV-nonrelated types include gastric, endometrioid, clear cell, mesonephric, serous and invasive adenocarcinoma NOS (Table 48.3).⁵⁶ The WHO updated its classification in 2020 to align with the IECC guidelines after demonstration of superior interobserver agreement and highly significant correlation with HPV status

TABLE 48.2

FIGO Staging of Cancer of the Cervix Uteri (2018)

Stage	Description
I	The carcinoma is strictly confined to the cervix (extension to the uterine corpus should be disregarded)
IA	Invasive carcinoma that can be diagnosed only by microscopy, with maximum depth of invasion <5 mm ^a
IA1	Measured stromal invasion <3 mm in depth
IA2	Measured stromal invasion >3 and <5 mm in depth
IB	Invasive carcinoma with measured deepest invasion >5 mm (greater than Stage IA); lesion limited to the cervix uteri with size measured by maximum tumor diameter ^b
IB1	Invasive carcinoma >5 mm depth of stromal invasion and <2 cm in greatest dimension
IB2	Invasive carcinoma >2 and <4 cm in greatest dimension
IB3	Invasive carcinoma >4 cm in greatest dimension
II	The carcinoma invades beyond the uterus, but has not extended onto the lower third of the vagina or to the pelvic wall
IIA	Involvement limited to the upper two-thirds of the vagina without parametrial involvement
IIA1	Invasive carcinoma <4 cm in greatest dimension
IIA2	Invasive carcinoma >4 cm in greatest dimension
IIB	With parametrial involvement but not up to the pelvic wall
III	The carcinoma involves the lower third of the vagina and/or extends to the pelvic wall and/or causes hydronephrosis or nonfunctioning kidney and/or involves pelvic and/or para-aortic lymph nodes
IIIA	The carcinoma involves the lower third of the vagina, with no extension to the pelvic wall
IIIB	Extension to the pelvic wall and/or hydronephrosis or nonfunctioning kidney (unless known to be due to another cause)
IIIC	Involvement of pelvic and/or para-aortic lymph nodes (including micrometastases) ^c , irrespective of tumor size and extent (with r and p notations) ^d
IIIC1	Pelvic lymph node metastasis only
IIIC2	Para-aortic lymph node metastasis
IV	The carcinoma has extended beyond the true pelvis or has involved (biopsy proven) the mucosa of the bladder or rectum. A bullous edema, as such, does not permit a case to be allotted to Stage IV
IVA	Spread of the growth to adjacent pelvic organs
IVB	Spread to distant organs

^aImaging and pathology can be used, where available, to supplement clinical findings with respect to tumor size and extent, in all stages. Pathological findings supersede imaging and clinical findings.

^bThe involvement of vascular/lymphatic spaces should not change the staging. The lateral extent of the lesion is no longer considered.

^cIsolated tumor cells do not change the stage but their presence should be recorded.

^dAdding notation of r (imaging) and p (pathology) to indicate the findings that are used to allocate the case to Stage IIIC. For example, if imaging indicates pelvic lymph node metastasis, the stage allocation would be Stage IIIC1r; if confirmed by pathological findings, it would be Stage IIIC1p. The type of imaging modality or pathology technique used should always be documented. When in doubt, the lower staging should be assigned.

From Bhatla N, Aoki D, Sharma DN, Sankaranarayanan R. Cancer of the cervix uteri: 2021 update. *Int J Gynaecol Obstet* 2021;155(S1):28–44. doi:10.1002/ijgo.13865.

TABLE 48.3

International Endocervical Adenocarcinoma Criteria and Classification

Main Categories	Morphologic Subcategories (Optional)	World Health Organization
1. HPV-related endocervical adenocarcinoma	HPV adenocarcinoma, usual type	1. Usual type adenocarcinoma
	HPV adenocarcinoma, villoglandular type	2. Villoglandular adenocarcinoma
	HPV adenocarcinoma, mucinous NOS type	3. Mucinous adenocarcinoma NOS
	HPV adenocarcinoma, mucinous intestinal type	4. Mucinous adenocarcinoma, intestinal type
	HPV adenocarcinoma, signet-ring cell type	5. Mucinous adenocarcinoma, signet ring cell
2. Non-HPV adenocarcinoma, gastric type	HPV adenocarcinoma, stratified mucin-producing type	6. Mucinous adenocarcinoma, gastric type
3. Non-HPV adenocarcinoma, endometrioid type		7. Endometrioid carcinoma
4. Non-HPV adenocarcinoma, clear cell type		8. Clear cell carcinoma
5. Non-HPV adenocarcinoma, mesonephric type		9. Mesonephric carcinoma
6. Non-HPV adenocarcinoma, serous type		10. Serous carcinoma
7. Invasive adenocarcinoma, NOS		11. Invasive adenocarcinoma, NOS

HPV, human papillomavirus; NOS, not otherwise specified.

With permission from Hodgson A, Park KJ, Djordjevic B, Howitt BE, Nucci MR, Oliva E, Stolnicu S, Xu B, Soslow RA, Parra-Herran C. International Endocervical Adenocarcinoma Criteria and Classification: Validation and Interobserver Reproducibility. *Am J Surg Pathol*. 2019;43(1):75–83.

using the IECC guidelines compared with the WHO 2014 guidelines, suggesting both greater biological and clinical significance.⁵⁷

Most endocervical adenocarcinomas are associated with HPV, commonly 16, 18, and 45; however, approximately 15% are HPV unrelated with distinct molecular alterations. The changes adopted in the 2020 WHO and 2018 IEACC guidelines reflect the distinct differences of HPV-related and HPV-unrelated adenocarcinomas. Using IECC system, if mitotic figures and apoptotic bodies are identified at low power, the tumor is likely HPV related, whereas difficulty locating mitotic figures or apoptotic bodies suggests an HPV-unrelated tumor. The most common HPV-related endocervical adenocarcinoma is the usual type (including villoglandular and micropapillary architectural variants), accounting for 75% to 80% of all endocervical adenocarcinomas. Cells of usual type typically are columnar, mucin-depleted cytoplasm, apical mitotic figures, and basal apoptotic bodies with pseudostratified, elongated, and hyperchromatic nuclei. Villoglandular architecture has prominent exophytic papillary growth on the superficial portion, with central fibrous cores, and when stromal invasion is present, it resembles the usual type; therefore, the villoglandular type was reclassified as an architectural variant in the 2020 classification rather than its own subtype. Another subtype is the micropapillary pattern with small papillary groups of neoplastic cells with eosinophilic cytoplasm and atypical nuclei surrounded by clear spaces and commonly associated with lymphovascular space invasion (LVSI) and metastatic disease. Mucinous (non-gastric-type) adenocarcinomas are a second group of HPV-related endocervical adenocarcinomas and are generally rarer. Similar to other HPV-related types, they display mitoses and apoptotic bodies at low-power magnification, and uniquely, the stroma may display mucin pools and inflammatory infiltrates, with a wide range of architecture. The signet-ring subtype is very rare and may indicate metastatic disease to the cervix. Invasive stratified mucinous carcinoma (ISMC) is characterized by invasive nests of stratified columnar cells with peripheral palisading and variable intracytoplasmic mucin and has been reported to have poorer prognosis than other HPV-related types. The ISMC subtype can display a wide range of architecture, including solid, papillary, micropapillary, trabecular, and others, along with variable cytologic appearance, including “glassy cell,” signet ring–like, cytoplasmic clearing, or “squamoid differentiation,” making diagnosis difficult.⁵⁷

Human papillomavirus–unrelated endocervical adenocarcinomas include gastric, clear cell, mesonephric, and endometrioid-type

carcinomas and adenocarcinoma NOS. The gastric type is the second most common adenocarcinoma of the cervix, with a relatively higher incidence in Southeast Asia. These tumors are characterized by basally located rounded nuclei and appear pale compared with the usual type with variable architecture and atypia. Minimal deviation adenocarcinoma (adenoma malignum) has a “claw-like” pattern of glands and has been associated with Peutz-Jeghers syndrome; however, this categorization is no longer recommended. Instead, *highly differentiated gastric-type adenocarcinoma* is used. Clear cell–type endocervical adenocarcinoma is very rare, typically seen in women exposed to diethylstilbestrol, and is characterized by polygonal or hobnail cells with abundant clear or eosinophilic cytoplasm and tubulocystic architecture. Mesonephric-type adenocarcinomas are exceedingly rare, originating from mesonephric remnants in the lateral cervical wall and exhibit a variety of growth patterns. The endometrioid type is thought to arise in the setting of endometriosis and is very rare; typically HPV negative; and typically estrogen receptor, progesterone receptor positive and p16 patchy, as opposed to the opposite in usual-type adenocarcinomas.⁵⁷

Small-cell/neuroendocrine carcinomas of the cervix resemble those of the lung and are characterized by small tumor cells with minimal cytoplasm, small round to oval nuclei, and high mitotic activity, and they typically display neuroendocrine features. Anaplastic small-cell carcinomas tend to behave more aggressively than poorly differentiated small-cell squamous carcinomas with poor survival rates even among those with early-stage disease.⁵⁸

CLINICAL MANIFESTATIONS

Preinvasive disease is usually detected during routine cervical cytologic screening. Early invasive disease may not be associated with any symptoms and is also usually detected during screening examinations. The earliest symptom of invasive cervical cancer is usually abnormal vaginal bleeding, often after intercourse. This may be associated with a clear or foul-smelling vaginal discharge. Pelvic pain may result from locoregionally invasive disease or from coexistent pelvic inflammatory disease. Flank pain may be a symptom of hydronephrosis, which may be complicated by pyelonephritis. Patients with advanced tumors may have hematuria or incontinence from a vesicovaginal fistula caused by direct extension of

tumor to the bladder. External compression of the rectum by a massive primary tumor may cause constipation, but the rectal mucosa is rarely involved at initial diagnosis.

CLINICAL EVALUATION AND STAGING

Clinical Evaluation of Patients with Invasive Carcinoma

All patients with invasive cervical cancer should be evaluated with a detailed history and physical examination, with particular attention paid to inspection and palpation of the pelvic organs with bimanual and rectovaginal examinations. The pattern of metastasis in carcinoma of cervix initially involves pelvic lymph nodes followed by the paraaortic nodes and distant sites. The most frequent metastatic sites are the lungs, extrapelvic nodes, liver, and bones. However, because of multidisciplinary treatment and increased survival, different metastatic sites have been reported, including the cervical lymph nodes.

Standard laboratory studies should include a complete blood cell count and renal function and liver function tests. There is increased use of cross-sectional imaging modalities (computed tomography [CT], magnetic resonance imaging [MRI], positron emission tomography/CT [PET/CT]) for the study of important prognostic factors such as tumor size, parametrial invasion, endocervical extension, pelvic side wall or adjacent or distal organ involvement, and lymph node status. MRI can be used to evaluate the size and extent of the cervical mass as well as suggest invasion into the parametria, bladder, or rectum. Many clinicians obtain CT or MRI scans to evaluate regional lymph nodes, but these studies have suboptimal accuracy because they fail to detect small metastases and patients with bulky necrotic tumors often have enlarged reactive lymph nodes that may be free of metastasis.

In a large Gynecologic Oncology Group (GOG) study that compared the results of radiographic studies with subsequent histologic findings, Heller et al.⁵⁹ found that the sensitivity of CT in the detection of positive paraaortic nodes was only 34%. PET appears to be a more sensitive, specific, and noninvasive method of evaluating the regional nodes of patients with cervical cancer.⁶⁰ Two phase II studies have further evaluated the sensitivity of fluorodeoxyglucose (FDG) PET/CT in the detection of pathologically positive paraaortic nodes. In these studies, all patients had stages IB2 to IVA cervical cancer and underwent PET/CT followed by surgical staging via removal of the paraaortic nodes using a laparoscopic extraperitoneal approach. In the first study, 60 evaluable patients successfully underwent the procedure. The overall sensitivity of PET in detecting metastatic disease to the paraaortic nodes when CT scan or MRI results were negative was only 36%. For women with completely negative PET scan results, 12% had disease spread to the nodes along the aorta.⁶¹ A similar study was performed in which 98 patients with locally advanced cervical cancer underwent PET followed by laparoscopic extraperitoneal paraaortic lymphadenectomy. For all patients, the sensitivity of PET/CT was 91.6% in detecting disease in the paraaortic region.⁶² The American College of Radiology Imaging Network (ACRIN) and the GOG (ACRIN 6671/GOG 0233) assessed the accuracy of staging PET/CT in detecting distant metastasis in patients with locoregionally advanced cervical and high-risk endometrial cancer. Central reader PET/CT interpretation demonstrated sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of 54.8%, 97.7%, 79.3%, and 93.1%, respectively, for cervical cancer.⁶³

Cystoscopy and proctoscopy can be considered in patients with bulky tumors and patients with imaging findings suggestive of organ involvement.

In the 2018 FIGO staging system for cervical cancer, imaging is formally incorporated as a source of staging information and as

a supplement to clinical examination (i.e., pelvic examination, cystoscopy, and colposcopy) to obtain an accurate description of tumor spread. The new staging system includes a more precise measurement of primary tumor size, best assessed with imaging, and introduces a new primary tumor size cutoff value of 2 cm (i.e., stage IB1 vs IB2) used to evaluate patients for fertility-sparing radical trachelectomy and to estimate prognosis. Evaluation for retroperitoneal lymphadenopathy in the abdomen and pelvis, also new to the 2018 FIGO revision, can be diagnosed at imaging alone or with pathologic analysis. Choice of modality depends on the technology available within the practice setting. In high-resource settings, pelvic MRI and PET/CT are used to assign stage. In lower resource settings, analogous modalities are pelvic ultrasonography and chest radiography.⁶⁴

Surgical Evaluation of Regional Spread

Information regarding lymph node involvement helps to guide whether the primary therapy will be radical hysterectomy or chemoradiation and if adjuvant chemoradiation should be given. Lymphadenectomy is performed as part of the surgical treatment of most patients with early-stage carcinomas of the cervix and is now part of FIGO staging. Pelvic and/or paraaortic lymph node dissection is occasionally used in selected patients with locally advanced-stage disease.

The use of sentinel lymph node (SLN) evaluation as a possible alternative to full lymphadenectomy in patients undergoing radical hysterectomy for cervical cancer is increasing, particularly among surgeons experienced with the procedure.⁶⁵ According to the SLN concept, tumor cells metastasize to one or a few lymph nodes before involving further lymph nodes. A tracer injected interstitially in the vicinity of the tumor is transported via local lymphatic channels toward the draining lymphatic basins, and the first node that the tracer reaches is called the “sentinel node.” The SLN mapping technique involves a process that allows the selective removal of few lymph nodes. Near-infrared technology and the use of indocyanine green (ICG) recently emerged as the dye of choice for SLN mapping. A noninferiority, randomized trial of 180 patients with uterine or cervical cancer found that, in total, 471 (97%) of 485 lymph nodes were identified with ICG and 226 (47%) with isosulfan blue.⁶⁵ It was established that ICG injected into the cervix should be considered the new standard of care for SLN mapping in women with cervical and uterine cancers.

Sentinel lymph node evaluation is usually reserved for patients with tumors smaller than 4 cm when no suspicious lymph nodes are identified during preoperative imaging. Ultrastaging is an integral part of the SLN evaluation. This procedure involves additional sectioning and staining of the SLN with hematoxylin and eosin stain and immunohistochemistry to examine the node for low-volume metastatic disease. Optimal histopathologic and/or molecular assessment protocols of nodal tissue remains unknown and important questions remain concerning the clinical management of patients with (very) low-volume disease.

Evaluation studies have appraised the high-sensitivity, high-NPV, and low false-negative rates for metastasis detection using SLN evaluation. In a meta-analysis of more than 3,900 patients from 44 studies, the sensitivity of SLNB for identifying nodal metastases was 81%; however, the sensitivity increased to 99% (95% confidence interval [CI], 98% to 100%) with an NPV of 97% to 100% when the analysis included only patients with tumors smaller than 4 cm in size and negative preoperative imaging results and in whom bilateral SLNs were detected intraoperatively with ultrastaging performed on sentinel node specimens.⁶⁶ Prospective studies have also confirmed the safety of this approach. In the prospective SENTICOL (Ganglion Sentinelle dans le Cancer du Col) study, the sensitivity and NPV of sentinel node biopsy in early cervical cancer were 92% and 98.2%, respectively.⁶⁷ In patients in whom

SLNs were identified bilaterally, sensitivity and NPV were both 100%. The 4-year follow-up on the SENTICOL 2 trial included 206 women across 25 centers over 3 years who were randomized to laparoscopic surgery with SLN alone or SLN combined with pelvic lymphadenectomy. There were similar disease-free survival (DFS) rates in women who were randomized to SLN alone (89.5%) versus women who underwent SLN and pelvic lymphadenectomy (93.1%) ($P = 0.53$).⁶⁸ The SENTIREC CERVIX (SENTInel node mapping in women with CERVical cancer) trial was a multicenter prospective cohort that included 245 women; 38 (15.5%) had nodal metastasis. The SLN detection rate was 96.3% (236 of 245), with 82.0% (201 of 245) bilateral detection. In a stratified analysis of 103 women with tumors larger than 20 mm, 27 (26.2%) had nodal metastases. The sensitivity of SLN mapping adhering to the algorithm was 96.3% (95% CI, 81.0%–99.9%), and the NPV was 98.7% (95% CI, 93.0%–100%). For FDG PET/CT imaging the sensitivity was 14.8%, the specificity was 85.5%, the NPV was 73.9% (95% CI, 63.4%–82.7%), and the PPV was 26.7% (95% CI, 7.8%–55.1%).⁶⁹

Although it is commonly accepted that FDG PET/CT performs poorly in the detection of microscopic disease spread to the paraaortic nodes in women with locally advanced cervical cancer, the role of surgical node evaluation before definitive radiation therapy of cervical cancer is a continued subject of controversy and rarely currently rarely perform. The evaluation procedure can be performed via laparotomy or laparoscopy through a transperitoneal or extraperitoneal approach. Extraperitoneal and laparoscopic approaches to staging (including extraperitoneal laparoscopic) are associated with reduced morbidity. In an effort to not undertreat the 16% to 32% of patients who have falsely negative PET scans for paraaortic disease, radiation oncologists may prophylactically extend the field to include these nodal basins. However, doing so is not without significant additional morbidity. The risk of grade 3 or 4 toxicity with extended field radiation approaches 15%. In 1989, Weiser et al.⁷⁰ reported that the proportion of patients with postradiotherapy bowel complications was reduced to less than 5% if lymphadenectomy was performed using a retroperitoneal approach. Because patients with radiographically positive pelvic nodes are at greatest risk for occult metastasis to paraaortic nodes, these patients may have the greatest chance of benefiting from surgical staging. Some investigators have advocated for resection of large pelvic nodes before radiotherapy to improve the rate of pelvic disease control.⁷¹ However, this approach has not been directly compared with modern conformal chemoradiotherapy, which can achieve high rates of disease control in grossly involved lymph nodes.⁷²

PROGNOSTIC FACTORS

Lymph node metastasis has long been considered one of the most important predictors of prognosis in patients with cervical cancer. Survival rates for patients treated with radical hysterectomy with or without postoperative radiotherapy for stage IB disease were usually reported as 85% to 95% for patients with negative nodes and 45% to 55% for those with lymph node metastases.⁷³ However, studies have demonstrated much higher survival rates of approximately 80% for patients with early-stage node-positive cervical cancer treated with posthysterectomy chemoradiotherapy,⁷⁴ suggesting that the use of tailored adjuvant treatment has narrowed the survival gap for patients with positive or negative nodes. Survival rates for patients with positive paraaortic nodes treated with extended-field radiotherapy range from 10% to 50% depending on the extent of pelvic disease and paraaortic lymph node involvement. Improved diagnostic methods such as PET and the advent of highly conformal radiation therapy techniques, along the introduction of cytotoxic chemotherapy as radiosensitizers, may have contributed to recent improvements in the outcomes of patients with extensive nodal metastases.⁷²

For patients treated with radical hysterectomy, other histologic parameters that have been associated with a poor prognosis are LVSI, deep stromal invasion, histologic type, and parametrial extension.^{73,75,76} Roman et al.⁷⁶ reported a correlation between the percentage of histopathologic sections containing LVSI and the incidence of lymph node metastases. Whether patients with squamous carcinomas have a better overall prognosis compared with those with adenocarcinomas of the cervix remains controversial.^{73,75,77} Although older studies suggest that adenocarcinoma compared with squamous carcinomas is associated with worse outcomes, a 2020 study showed that when adjusted for stage, prognosis is similar. In this retrospective study of 4,131 patients with either adenocarcinoma or squamous carcinomas of the cervix from the Surveillance, Epidemiology, and End Results database, 5-year overall survival (OS) was similar between groups.⁷⁸ Although the use of concurrent chemotherapy with radiation appears to confer a benefit for both squamous carcinomas and adenocarcinomas,⁷⁹ patients who received chemoradiation for adenocarcinomas still tend to have a poorer prognosis than those with squamous cancers who receive similar treatment.⁸⁰ For patients with adenocarcinoma of the cervix, outcome appears to be correlated with the degree of tumor differentiation.⁸¹

Other clinical and biologic features have been reported to correlate with outcome, but none of these have been incorporated into routine clinical practice.⁸² Molecular markers such as tumor vascularity, cyclooxygenase-2 expression, galectin expression, and other growth factor receptors have been reported to correlate with outcome.^{83–86} Several investigators have found the presence of HPV DNA in histologically cancer-free lymph nodes to be correlated with poor outcome.⁸⁷ Tumor *PIK3CA* mutation status has also been identified as a prognostic factor in cervical cancer. McIntyre et al.⁸⁸ reported finding exon 9 or exon 20 *PIK3CA* mutations in 23% of the 84 cervical cancers studied; the authors found a strong correlation between *PIK3CA* mutation status in patients with stage IB or II carcinomas but not in those with stage III or IVA cancers. The Cancer Genome Atlas identified *SHKBP1*, *ERBB3*, *CASP8*, *HLA-A*, and *TGFBR2* as significantly mutated genes in cervical cancer. Lower OS rates were seen among patients whose tumors had a protein expression signature characterized by increased epithelial–mesenchymal transition (EMT); EMT is a process that has been linked with the formation of metastasis.⁸⁹

There is also some evidence that functional imaging parameters may be of prognostic value. Kidd et al.⁹⁰ concluded that the addition of standardized uptake values from the primary site of cervical cancers to other PET parameters (tumor size and the extent of lymph node involvement) improved their ability to predict the outcomes after chemoradiotherapy. In a study of 238 patients with cervical cancer treated with chemoradiation, all patients had PET and posttherapy PET 8 weeks to 16 weeks after treatment completion. The rates of recurrence among those with complete metabolic response, partial metabolic response, and progressive disease were 23%, 65%, and 100%, respectively, differences that were statistically significant. Although most failures were distant, a partial metabolic response predicted for locoregional failure.⁹¹

CHEMOTHERAPY FOLLOWED BY RADICAL SURGERY

A number of researchers have investigated the use of neoadjuvant chemotherapy followed by radical hysterectomy to treat patients with bulky stage IB or stage II cervical carcinoma.

Several early trials by Sardi and colleagues⁹² suggested better projected 4-year DFS rates when neoadjuvant chemotherapy was added to radical hysterectomy plus postoperative radiotherapy in patients whose tumors were larger than 4 cm. However, in a subsequent randomized trial,⁹³ the GOG reported no significant difference in recurrence rates (relative risk, 0.99) or death rates (relative risk, 1.008) for patients who did or did not receive neoadjuvant

chemotherapy before radical hysterectomy. In this trial, patients who underwent hysterectomy were treated with postoperative irradiation if they had high-risk disease features; the proportion requiring postoperative irradiation was similar in the two arms (45% for those who did and 52% for those who did not receive neoadjuvant chemotherapy).

In a single-center, phase III, randomized controlled trial, 635 patients with stage IB2, IIA, or IIB squamous cervical cancer were randomly assigned to either three cycles of neoadjuvant chemotherapy (paclitaxel and carboplatin once every three weeks) followed by radical hysterectomy or to standard chemoradiation. Compared with standard chemoradiation, those receiving neoadjuvant chemotherapy followed by surgery experienced worse five-year DFS (69.3% vs 76.7%, respectively; hazard ratio [HR] 1.38; 95% CI, 1.02 to 1.87). However, 5-year OS was similar between the two groups (75.4% vs 74.7%, respectively; HR, 1.025; 95% CI, 0.75–1.40). The delayed toxicities at 24 months or later after treatment completion in the neoadjuvant chemotherapy plus surgery group versus the concomitant chemoradiation group were rectal (2.2% vs 3.5%, respectively), bladder (1.6% vs 3.5%, respectively), and vaginal (12.0% vs 25.6%, respectively).⁹⁴ In preliminary results of a multicenter phase III trial, among 620 patients with stage IB2 to IIB cervical cancer, there was no difference in OS between those randomly assigned to neoadjuvant chemotherapy followed by surgery experienced comparable compared with those receiving concomitant chemoradiation (5-year OS, 72% vs 76%; HR, 0.87; 95% CI, 0.65 to 0.15).⁹⁵ Although neoadjuvant chemotherapy is rarely used in the United States, this later study suggests a possible role of this approach in some clinical scenarios.

There are promising but limited data regarding administration of neoadjuvant chemotherapy followed by fertility-sparing surgery in premenopausal women with tumors measuring larger than 2 cm to less than 4 cm who wish to preserve their childbearing potential.^{96,97} The Cervical Cancer Treated with Neoadjuvant Chemotherapy Followed by Fertility-sparing Surgery (CoNteSSa) trial (NCT04016389),⁹⁸ a single-arm multicenter phase II study, will provide further information regarding the safety of this strategy.

MANAGEMENT OF ABNORMAL CERVICAL SCREENING

The current guidelines from the ASCCP in 2019 base recommendations for management on 5-year risk of progression to CIN 3+ (CIN 3, AIS or invasive disease) as determined by the results of current screening tests and the patient's history. This is a change from prior result-based management. Recognition of the central role of persistent HPV infection in the development of precancer and cancer underlines these updated guidelines. Prospective longitudinal data demonstrate that risk of CIN 3+ decreases more substantially after a negative HPV test or co-test in the past 5 years compared with a negative cytology test result in the past 3 years. The updated guidelines were determined using data from the Kaiser Permanente of Northern California study of more than 1.5 million patients screened from 2003 to 2017 to produce risk profiles of CIN 2+, CIN 3+, and cancer for yearly time points from the time of the current test (immediate risk) to 5 years after the test (5-year risk). Risk-based thresholds were established for management, including expedited treatment for immediate CIN 3+ risk of 60% or greater, treatment or colposcopy or biopsy if immediate CIN 3+ risk of less than 60% and 25% or greater, colposcopy or biopsy recommended if immediate CIN 3+ risk is 4% or greater but less than 25%, retest in 1 year if 5-year CIN 3+ risk is 0.55% or greater but immediate risk less than 4%, retest in 3 years if 5-year CIN 3+ risk at least 0.15% but less than 0.55%, and retest in 5 years if 5-year CIN 3+ risk less than 0.15%. Notably, these risk thresholds were consensus decisions specific to the United States.⁹⁹ In general, updated guidelines have new principles of HPV-based testing for risk estimation (including primary HPV testing or co-testing

with cervical cytology), understanding of patient's past results, and maximizing colposcopic detection of CIN 2+ to avoid missing dysplasia and build on prior principles of the primary goal to prevent cancer among all individuals with a cervix, balancing the benefits and harms and specific application of guidelines to asymptomatic individuals undergoing screening in the United States⁴⁸ (Fig. 48.3).

Using the risk-based approach, CIN 3 is considered sufficient risk to necessitate treatment for all nonpregnant individuals. Excisional therapy includes LEEP, cold knife conization, or laser cone biopsy, whereas ablation treatment includes cryotherapy, laser ablation, and thermoablation.¹⁰⁰ Excisional treatment is the preferred method of treatment in the United States, though more adverse events are noted with cone and LEEP procedures than cryotherapy.⁴⁸ Observation of CIN 2 may be acceptable pending individual risk of CIN 3 for patients concerned about effects of treatment on future pregnancy outcomes.⁴⁸ For patients with histologic CIN 1 (LSIL) or less preceded by HSIL cytology, immediate excisional procedure or 1-year HPV-based testing with colposcopy is acceptable provided that colposcopic examination was sufficient, and for those preceded by ASC-H, 1-year HPV-based testing is recommended. If test results at 1 year are normal, repeat HPV-testing the following year is recommended before returning for testing every 3 years. Those with CIN 1 (LSIL) diagnosed repeatedly for 2 years are recommended to have excisional or ablative treatments. Patients with HPV+ ASC-US have similar risks of CIN 3+ as those with LSIL cytology and are thus managed similarly. For patients with AIS or AGC, colposcopy is recommended, with endometrial sampling for those over 35 years, or, if younger with additional risk factors for endometrial neoplasia. If histologic findings demonstrate AIS or AGC, excisional procedure is recommended, with a length of at least 10 mm and up to 20 mm for those unconcerned about risk to future pregnancy. After the initial diagnostic procedure, if AIS remains the diagnosis, hysterectomy is the preferred management.⁴⁸ Overall, the updated guidelines recommend reflex cytology and HPV genotyping for all positive HPV tests and longer duration of 3-year interval screening for any history of histologic HSIL, CIN 2, CIN 3, or AIS (at least 25 years).⁴⁸

TREATMENT OF INVASIVE DISEASE

There are several factors that influence decisions regarding treatment cervical cancer, including the depth of invasion, stage, histologic features, evidence of metastases to lymph nodes or beyond, risk factors for complications of surgery or radiation, and patient preference. In general, patients with microinvasive cancers (stage IA1) are treated with conservative surgery, patients with early invasive cancers (stage IA2 and IB1) have potential for a range of treatments from conservative surgery to simple or radical hysterectomy or radiotherapy, patients with stage IB2 and IIA cancers may be managed with radical hysterectomy, and patients with locally advanced cancers stage IB2 through IV cancers may be managed with chemotherapy and radiation. Patients with radiographically positive lymph nodes are typically treated with chemoradiation. Nodal metastases are most common in the external and internal iliac distribution followed by common and paraortic lymph node involvement (Fig. 48.4). For select patients with centrally recurrent disease after maximum radiotherapy, radical exenterative surgery may be considered.

Microinvasive Carcinoma (Stage IA)

Current recommended treatment for clinical FIGO stage IA1 without LVSI is to perform an excisional biopsy to evaluate for further invasive disease. A cone biopsy is preferred, though LEEP may be acceptable as long as orientation is maintained, and endocervical curetting is added as clinically indicated. If margins are negative, extrafascial hysterectomy is recommended. The risk of pelvic lymph node metastases in these tumors is less than 1%; therefore, pelvic lymphadenectomy is not recommended.¹⁰¹

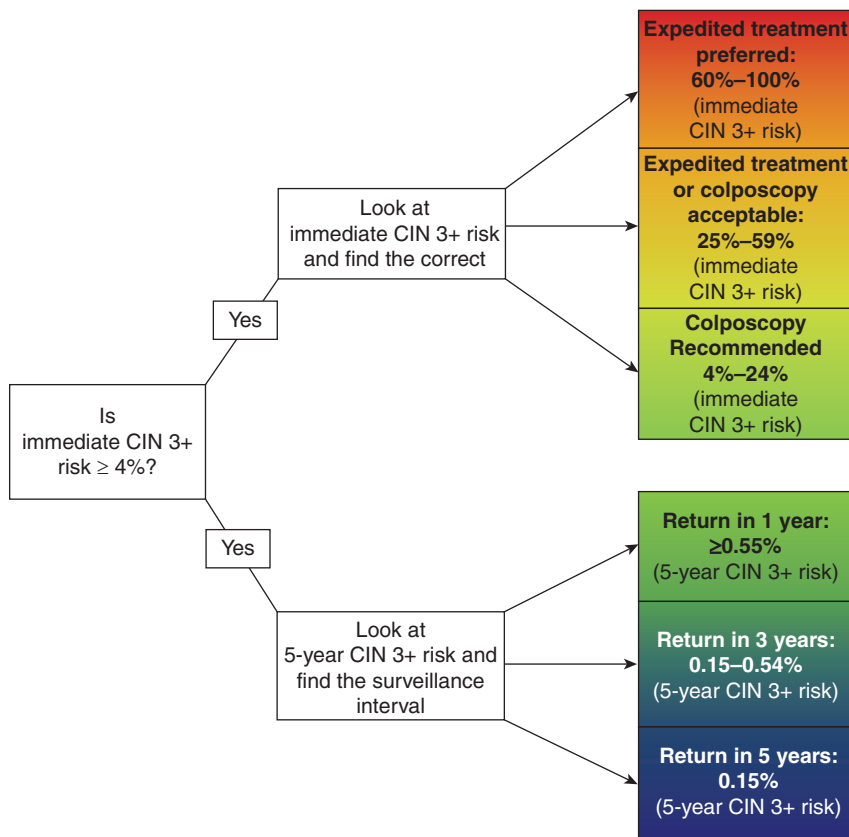


Figure 48.3 Evaluating patient risk in patients with dysplasia. For a given current results and history combination, the immediate cervical intraepithelial neoplasia (CIN) 3+ risk is examined. If this risk is 4% or greater, immediate management via colposcopy or treatment is indicated. If the immediate risk is less than 4%, the 5-year CIN3+ risk is examined to determine whether the patient should return in 1, 3, or 5 years. (Perkins RB, Guido RS, Castle PE, et al. 2019 ASCCP risk-based management consensus guidelines for abnormal cervical cancer screening tests and cancer precursors. *Low Genit Tract Dis* 2020;24(2):102–131.)

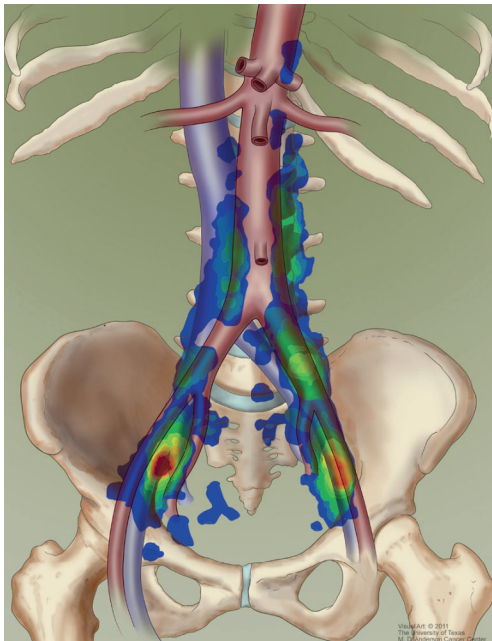


Figure 48.4 Anatomic distribution of positron emission tomography-positive lymph nodes based on a volume probability map. A color gradient corresponding to the visible-light spectrum is used to indicate the frequency of lymph node involvement. *Red*, high frequency; *green*, moderate frequency; *blue*, low frequency. (© 2011 the University of Texas MD Anderson Cancer Center. Copyright used with the permission of The Board of Regents of the University of Texas System through The University of Texas MD Anderson Cancer Center.)

For women who desire fertility-sparing treatments, those with clinical stage Ia1 without LVSI may undergo cone biopsy and if margins are negative may be monitored with close surveillance. Per consensus guidelines, if the margins are positive, a repeat excision or trachelectomy may be performed.¹⁰² Prior studies suggest risk of recurrence is infrequent¹⁰³; however, patients with this treatment should be followed closely with colposcopy, endocervical sampling, and HPV testing.

The risk of residual invasive disease following excisional cone biopsy is closely related to the status of the internal cone margin and results of endocervical curettage (ECC) performed after the biopsy.¹⁰⁴ Phongsarism et al.¹⁰⁴ found that depth of stromal invasion more than 1 mm in the cone specimen ($P = 0.014$), presence of invasive carcinoma at the margins ($P = 0.003$), and positive ECC ($P = 0.001$) were significantly associated with residual micro-invasive or frankly invasive carcinomas in hysterectomy specimens in a study of 129 patients. This study agreed with prior studies that recommended reexcision for those with residual disease (CIN or carcinoma) at the margins or positive ECC findings.¹⁰⁵

Therapeutic cone biopsy is performed using a scalpel with the patient under anesthesia. Because of the critical nature of the depth of invasion, the entire specimen should be handled carefully to maintain the original orientation. The depth and width of the cone should be tailored to provide clear surgical margins and produce the least amount of injury. Complications of cone biopsy include hemorrhage, sepsis, infertility, cervical stenosis, cervical incompetence, preterm birth or spontaneous abortion, and overall higher rates of complications than LEEP or ablative techniques.^{106–108} A current systematic review and metaanalysis is currently in process to compare efficacy and complication rates after local treatment of patients with CIN and stage Ia1 cervical cancer using cold knife, laser or Fisher cone, large loop, or needle excision versus ablative techniques.¹⁰⁹

The use of LVSI as a prognostic factor for survival in early-stage disease is limited by a small number of studies; however, LVSI is generally considered a high-risk pathologic characteristic and has been associated with higher rates of lymph node metastases in

early stage disease.^{110,111} Rates of LVSI in stage IA1 cervical cancers range from 3% to 22% and for IA2 from 11% to 43% among limited prior studies, perhaps partly because of wide interobserver variability in the diagnosis of LVSI. Margolis et al.¹⁰¹ found that LVSI increased the risk of nodal disease from less than 1% to 7.8% among women with stage IA1 tumors and from 1.7% to 14.6% among those with stage IA2 disease among a cohort of more than 4,000 women. Given the increased risk of nodal metastases with LVSI as well as the increased depth of invasion, patients with clinical stage IA1 with LVSI and stage IA2 are recommended to undergo evaluation of lymph nodes.¹⁰²

Early-Stage Cancer

Historically, patients with stage IA2 and IB1 and some IIA tumors were treated with radical hysterectomy and removal of pelvic lymph nodes. In the United States, consensus guidelines recommend modified radical hysterectomy (Type II) for patients with stage IA1 with LVSI and IA2 recommend for patients with stage IA1 with LVSI and IA2, whereas classical radical hysterectomy (type III) is recommended for those with stage IB1 (>2 cm) disease IB2 and some cases of IB3 to IIA1 (Table 48.4). For patients with stage IB1 squamous carcinomas, surgical treatment is usually preferred for healthy women with small tumors, particularly for premenopausal women in whom ovarian preservation may occur. Radiotherapy with chemotherapy is an option for older women and those with significant medical comorbidities.

Table 48.4 and Figure 48.5 provide a comparison of hysterectomy types as well as fertility-sparing trachelectomy types. Compared with an extrafascial hysterectomy, modified and radical hysterectomies involve removal of the paracervical, parametrial, and paravaginal tissues as well as uterosacral and portions of the cardinal ligaments, as well as the proximal vagina. A type III radical hysterectomy involves more extensive resection of the surrounding tissues, ligation of the uterine vessels at their origin, and resection of the upper third of the vagina. In a modified radical hysterectomy, the uterus, cervix, upper vagina, paracervical tissues, medial halves of the cardinal ligament, and uterosacral ligaments are removed, with careful dissection of the ureters to the bladder. The ovaries may or may not be removed, depending on the individual patient's age, menopausal status, and other factors, and may be transposed out of the pelvis if postoperative radiation is believed to be necessary because of intraoperative findings (Table 38.4).

With the rise in minimally invasive surgery (MIS), radical hysterectomy was performed using MIS techniques, and observational studies appeared to demonstrate equivalent outcomes; however, recent investigations have demonstrated that minimally invasive radical hysterectomy is associated with lower rates of disease-free and OS than open abdominal hysterectomy. The phase III Laparoscopic Approach to Cervical Cancer (LACC) trial compared 319 patients undergoing MIS and 312 patients undergoing open radical hysterectomy and found a lower rate of 3-year disease free survival (91.2% vs 97.1%) and OS (93.8% vs 99.0%) in the MIS arm than the open surgery arm.¹¹³ The results of this trial were

TABLE 48.4

Principles of Evaluation and Surgical Staging: Resection of Cervical Cancer as Primary Therapy

Comparison of Hysterectomy Types				Comparison of Fertility-Sparing Trachelectomy Types	
	Extrafascial Hysterectomy (Type A) ^b	Modified Radical Hysterectomy (Type B) ^b	Radical Hysterectomy (Type C) ^b	Simple Trachelectomy	Radical Trachelectomy ^c
Indication	Stage IA1	Stage IA1 with LVSI and IA2	Local disease without obvious metastasis, including stages IB1–IB2 and selected stages IB3–IIA1	Carcinoma in situ and stage IA1	Stage IA2–IB1 Select IB2
Intent	Curative for microinvasion	Curative for small lesions	Curative for larger lesions	Curative for microinvasion Fertility preserved	Curative for select stages IA2–IB2 Fertility preserved
Uterus	Removed	Removed	Removed	Spared	Spared
Ovaries	Optional removal	Optional removal	Optional removal	Spared	Spared
Cervix	Completely removed	Completely removed	Completely removed	Majority removed (~5 mm of the cranial aspect of the cervix typically left for cerclage)	Majority removed (~5 mm of the cranial aspect of the cervix typically left for cerclage)
Vaginal margin	Minimal	1–2 cm margin	Upper 1/4 to 1/3 of vagina	Minimal	1- to 2-cm margin
Ureteral dissection	Not mobilized	Ureters unroofed and dissected from cervix	Ureters unroofed and dissected from cervix and from lateral parametria	Not mobilized	Ureters unroofed and dissected from cervix

^aCibula D, Abu-Rustum NR, Benedetti-Panici P, et al. New classification system of radical hysterectomy: Emphasis on a three-dimensional anatomic template for parametrial resection. *Gynecol Oncol* 2011;122:264–268.

^bQuerleu D, Morrow CP. Classification of radical hysterectomy. *Lancet Oncol* 2008;9:297–303.

^cFertility-sparing radical trachelectomy is most validated for lesions ≤2 cm in diameter. Small-cell neuroendocrine histology and gastric type adenocarcinoma are not considered suitable tumors for this procedure.

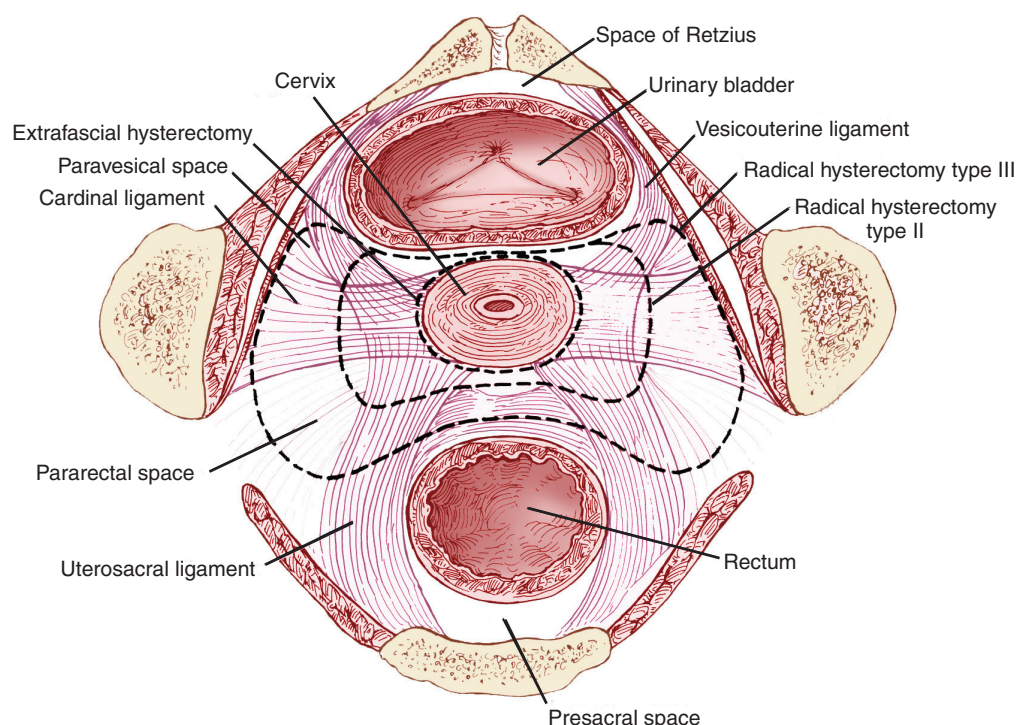


Figure 48.5 The pelvic ligaments and spaces. Dotted lines indicate the tissues removed with a modified radical (type II) or radical (type III) hysterectomy.

corroborated by epidemiologic studies. Given these findings, current recommendation is for an open approach to radical surgery and careful discussion with patients regarding the oncologic risks of alternative approaches.^{113,114}

Intraoperative and immediate postoperative complications of radical abdominal hysterectomy include blood loss, ureterovaginal fistula (1% to 2% of patients), vesicovaginal fistula (<1%), pulmonary embolus (1% to 2%), small bowel obstruction (1% to 2%), and postoperative fever secondary to deep vein thrombosis, pulmonary infection, pelvic cellulitis, urinary tract infection, or wound infection (25% to 50%).¹¹⁵ Complications of pelvic lymphadenectomy performed during radical surgery include lymphocyst formation, which may cause mass effect on the ureter, and lower extremity edema related to the extent of the lymphadenectomy.¹¹⁶ The risk of complications, particularly small bowel obstruction, may additionally be increased among patients who undergo adjuvant radiation.¹¹⁷

Most patients have transiently decreased bladder sensation after a radical hysterectomy; however, using a modified approach, significant urinary tract and other complications are minimized.¹¹⁸ With careful postoperative bladder drainage, chronic bladder hypotonia persists in 3% to 5% of patients; however, severe long-term bladder complications are infrequent and are typically related to the extent of the parametrial and paravaginal dissections.¹¹⁹

Early-Stage Cancer: Fertility Conservation

In the past decade, there has been increased attention paid to the potential for fertility-conserving surgery for patients with low-risk features and stage IA2 to IB1 tumors. Radical trachelectomy with pelvic lymph node assessment has been demonstrated to have equivalent oncologic outcomes, with removal of the cervix, upper vagina, and parametrium and sparing of the uterine fundus to allow for future pregnancy.¹²⁰ The procedure also involves the placement of a nonabsorbable cervical cerclage around the uterine isthmus at the time of trachelectomy to keep the residual uterine segment intact. Successful pregnancies have been reported in a significant percentage of patients after radical

trachelectomy.¹²¹ Among patients with radical trachelectomy, there were increased rates of dysmenorrhea, irregular menstruation, cervical suture problems, stenosis, and prolonged amenorrhea compared with patients who underwent radical hysterectomy.¹²² Further studies are necessary to guide appropriate surgical approach for radical trachelectomies, with initial analyses demonstrating equivalent survival in minimally invasive or open approaches, though these are limited by small numbers of outcomes of interest.¹²³ Many institutions have adopted an open approach following the LACC trial.

Despite excellent local control of disease, radical hysterectomies and trachelectomies are associated with significant morbidity because of removal of nerve fibers associated bladder, bowel, and sexual function, in addition to surgical complications such as hemorrhage, injury to bladder or ureter, and fistula formation.¹²⁴ Recently, the utility of resection of the parametrium in women with early-stage disease and favorable pathologic findings (tumor size ≤ 2 cm, depth of invasion ≤ 10 mm, and negative pelvic nodes) has come into question. Retrospective studies have reported that fewer than 1% of these patients have parametrial involvement, and small prospective studies demonstrated favorable results in select women with low-risk cancer undergoing conization or simple hysterectomy with lymph node assessment.^{125–130}

The ConCerv Trial, a prospective, single-arm multicenter study from 2010 to 2019 to evaluate feasibility and oncologic outcomes, evaluated 100 women with IA2 to IB1 cervical carcinoma with low-risk features who underwent conservative management with conization and lymph node assessment or conization followed by simple hysterectomy with lymph node assessment. The trial demonstrated that conservative surgery with conization and simple hysterectomy is feasible in those with early-stage, low-risk carcinoma, with a positive lymph node rate of 5% and rate of residual disease in the hysterectomy specimen after conization of 2.5%.¹²⁴ Two additional prospective trials are ongoing, the Radical versus Simple Hysterectomy and Pelvic Node Dissection with Low-Risk Early-Stage Cervical Cancer (SHAPE) Trial and the GOG 278 trial, to evaluate conservative surgery in low-risk cervical cancer.

The result of the ConCerv Trial should be interpreted in light of limitations and applicability to a narrow subset of patients; however, results from this and other prospective studies suggest a role for less invasive procedures in treating patients with early-stage low-risk cervical cancer.

Surgical treatment remains the standard for in situ and micro-invasive cancer; however, for a subset of patients with severe medical comorbidities or other contraindications to surgical treatment, radiotherapy can provide successful treatment. Depending on the depth of invasion, LVSI, and other clinical features, patients with early-stage cervical cancer may be managed with brachytherapy and external beam radiation.¹³¹

Stage IB and IIA without Fertility Preservation

Patients with early-stage IB (IB1, IB2) and stage IIA1 cervical carcinomas may be treated effectively with combined external-beam irradiation and brachytherapy with or without concurrent platinum chemotherapy or with radical hysterectomy and bilateral pelvic lymphadenectomy. The goal of treatment is destruction of malignant cells of the cervix, paracervical tissues, and regional lymph nodes.¹¹⁷ For those who are treated with radical hysterectomy whose tumors have poor prognostic factors, postoperative radiotherapy or chemoradiation may be considered.¹³²

Some studies have suggested radical hysterectomy as the initial treatment for select patients with IB3 tumors; however, those with tumors measuring more than 4 cm often require adjuvant radiation or chemoradiation,⁷⁴ thus increasing the duration of exposure to treatment and potential side effects. Those with pathological high-risk features after radical surgery such as parametrial invasion, positive margins, and lymph node metastases are generally recommended to have adjuvant therapy (either radiotherapy with concurrent cisplatin or cisplatin-based chemotherapy).^{133–135} However, there is not current consensus in the role of chemotherapy in addition to radiotherapy for those with intermediate-risk factors (i.e., large tumor size, deep invasion, or LVSI).^{133,134,136} A phase III study of adjuvant radiotherapy, concurrent chemoradiation, or sequential chemoradiation in 1048 women with stage IB to IIA cervical cancer demonstrated sequential chemoradiation resulted in higher disease free survival and lower risk of cancer death than radiotherapy or concurrent chemoradiation, with no difference in disease free survival or cancer death between radiation and concurrent radiation.¹³⁶ For bulky (>4 cm) low-stage tumors, there may be a role for preoperative concurrent radiation and chemotherapy followed by surgery in select patients.¹³⁵

One large, randomized trial compared initial surgical management with definitive radiation therapy for patients with stage IB or IIA cervical cancer. In that trial, published by Landoni et al.¹¹⁷ in 1997, 343 women were randomized to radical (type III) hysterectomy or a combination of external-beam and low-dose-rate intracavitary brachytherapy. Patients did not receive concurrent chemotherapy with their radiation and had a low median dose to point A of 76 Gy, and 54% of those in the surgical arm required postoperative pelvic radiation for those with tumors 4 cm or smaller and 84% of those with tumors larger than 4 cm. The study authors concluded there were no differences in 5-year survival rates between surgical and radiotherapy intervention (80% and 82% respectively, for those with tumors ≤4 cm and 63% and 57% for those with tumors >4 cm).¹¹⁷ A 20-year update of this initial study was conducted, demonstrating OS of 72% in the surgical group and 77% in the radiotherapy group ($P = 0.28$) and confirmed risk factors for survival were related to histotype, tumor diameter, and lymph node status, not treatment modality.¹³⁷ Notably, in the 20-year interval, retrospective database reviews suggested advantage for surgical treatment; however, these were limited by significant selection bias.¹³⁷

Treatment Following Simple Hysterectomy with Unsuspected Invasive Cancer

Before undergoing hysterectomy, careful screening should be conducted to rule out invasive cervical cancer. Despite screening, cervical cancer may be found incidentally, with some estimates of occurrence up to 5% of cases.¹³⁸ When unexpected invasive cancer is detected, patients should be considered for additional treatment. For those with IA1, further treatment is not necessary, with some studies suggesting no additional treatment even in the case of LVSI.¹³⁸ For more advanced early-stage tumors (stage IA2–IIA), definitive treatment with adjunctive radical parametrectomy, radiotherapy or concurrent chemoradiation is recommended. Park et al.¹³⁸ retrospectively evaluated outcomes in 147 patients with occult invasive cancer, 99 of whom were stage IA2–IIA who were observed ($n = 26$), received radiation or chemoradiation ($n = 44$), or underwent radical parametrectomy ($n = 29$). Overall radical parametrectomy and radiation and chemoradiation had similar efficacy with lower rates of recurrent disease; however, late complications were higher among those who received chemotherapy, suggesting radical parametrectomy may be preferable.¹³⁸ In the United States, consensus guidelines suggest for stage IA1 with LVSI, stage IA2/IB with positive margins, pelvic external beam radiation with concurrent chemotherapy with or without brachytherapy, and if negative margins consider completion of parametrectomy, upper vaginectomy and pelvic lymphadenectomy if Sedlis criteria were not met on the hysterectomy specimen.¹⁰²

Carcinoma of the Cervical Stump

Supracervical hysterectomy is an alternative treatment to total hysterectomy that leaves the cervix in situ and is performed for a variety of reasons, including patient preference, surgeon preference, and perceived operative difficulty of removal of the cervix, but has not been demonstrated to be superior in patient-reported outcomes of urinary or sexual dysfunction, pelvic floor instability, or clinically significant perioperative and postoperative complications and leaves the patient at risk for development of cervical carcinoma in the cervical stump.¹³⁹ It is overall a rare presentation, accounting for 2% to 5% of all cervical cancer cases worldwide diagnosed since 1970.¹⁴⁰ The natural history, staging, and workup for cervical stump carcinomas are the same for those with an intact uterus. Those with stage IA1 disease may be treated with simple trachelectomy, and select patients with stage IA2 or small stage IB tumors may be treated with radical trachelectomy and pelvic lymphadenectomy. Those with more advanced disease are typically treated with chemoradiation and a combination of external-beam radiation and brachytherapy.¹⁴⁰ In review of 203 cases, Barillot et al.,¹⁴¹ reported a survival rate of 81.5% of patients treated with combined brachytherapy and external-beam radiation versus 38.5% with external-beam radiation alone. Given the altered geometry after supracervical hysterectomy, MRI may assist with planning for brachytherapy; however, in most cases, the endocervical canal remains 2 cm or more after supracervical hysterectomy, allowing for intracavitary therapy or at least interstitial therapy.^{142,143}

Carcinoma of the Cervix During Pregnancy

Cervical cancer is the most common gynecologic tumor diagnosed during pregnancy, though the incidence is overall low, with an estimate of 0.8 to 1.5 cases per 10,000 births. Approximately 1% to 3% of women who are diagnosed with cervical cancer are pregnant or postpartum, with approximately 50% of these cases diagnosed during prenatal care.¹⁴⁴

Any suspicious cervical lesion observed during pregnancy should be biopsied. Colposcopic or naked eye biopsies may be taken during pregnancy without increased risk of complications during the pregnancy; however, ECC and endometrial sampling are contraindicated in pregnancy because of increased rates of abortion or other complications.¹⁴⁴ Conization may be undertaken in pregnancy; however, it is typically recommended only in the second trimester because of increased rates of abortion in the first trimester ($\leq 33\%$) and only in patients for whom invasive cancer is strongly suspected with inadequate visualization of the squamocolumnar junction on colposcopy.¹⁴⁴ For biopsy-proven invasive cervical cancer, staging workup is indicated. For those with suspected IB1 cancer or greater, chest radiography for pulmonary metastases and MRI for radiologic staging are recommended.¹⁴⁵ Some studies have reported lymphadenectomy up to 20 weeks' gestation without adverse outcomes.¹⁴⁶ A discussion with the patient regarding the option to terminate the pregnancy if diagnosis is made before viability, and of chemoradiation-inducing abortion is considered if the diagnosis is made at an early gestational age.¹⁴⁶

It appears safe to delay definitive treatment in patients with carcinoma in situ or stage IA disease until fetal maturity, and those with invasion of less than 5 mm may be followed to term. Although radical trachelectomies have been performed for patients with stage IB1 and IB2, the rate of fetal loss after this procedure is very high, such that it should only be considered in centers with extensive experience and after thorough counseling of the patient. Those with IB1 and negative nodes may delay definitive treatment to fetal maturation, at which point cesarean section followed by radical hysterectomy may be considered. Those with disease beyond IB1 who desire to continue a pregnancy (if before viability) or are at a gestational age less than fetal maturity are recommended to consider neoadjuvant chemotherapy, with radiation, surgery, or both after delivery.¹⁴⁶ The recommended route of delivery is via cesarean section for those with known cervical malignancy to decrease rates of bleeding, recurrence in episiotomy sites or rare vertical transmission of malignancy.^{147–149} An analysis of cesarean radical hysterectomies in the United States found high surgical morbidity (43%–45%) with high blood loss but no difference in survival between open radical hysterectomy and cesarean radical hysterectomy.¹⁵⁰ If the diagnosis of malignancy is made after fetal maturity, the recommendation is delivery to not delay initiation of oncologic treatment.

Overall, compared with other patients with cervical cancer, those diagnosed during pregnancy do not appear to have worse survival outcomes and may be diagnosed at an earlier stage because of increased surveillance during pregnancy.¹⁵¹

Radiotherapy After Radical Hysterectomy

Radiation treatment decreases the risk of pelvic recurrence after radical hysterectomy in patients with high-risk disease features (lymph node metastasis, deep stromal invasion, positive or close operative margins, or parametrial involvement). GOG-92¹⁵² evaluated the benefit of adjuvant pelvic irradiation in patients with an intermediate risk of recurrence after radical hysterectomy for stage IB carcinoma. Patients were eligible for this study if they had a combination of risk factors, including greater than one-third stromal invasion, LVSI, or clinical tumor diameter of at least 4 cm. Patients with involvement of the pelvic lymph nodes, parametria, or surgical margins were excluded. Patients who received adjuvant radiotherapy experienced a 46% reduction in the risk of recurrence ($P = 0.007$). There was a 30% reduction in the risk of death for patients who received radiotherapy ($P = 0.07$). A subset analysis suggested that the benefit of postoperative radiotherapy was particularly striking for patients who had adenocarcinomas or adenosquamous carcinomas.^{132,153}

Although pelvic irradiation reduces the risk of recurrence for patients with pelvic lymph node metastases or parametrial involvement, the risk of pelvic and distant recurrence remains high for these women after radiotherapy. To improve the results

of combined-modality treatment, the Southwest Oncology Group (SWOG) conducted a prospective trial comparing postoperative pelvic radiotherapy alone versus administration of cisplatin and 5-fluorouracil (5-FU) during and after postoperative pelvic radiotherapy for patients with lymph node metastases, parametrial involvement, or involved surgical margins. Initial results of this trial, published in 2000, demonstrated significantly improved rates of pelvic disease control and survival for patients who received chemotherapy in addition to radiation treatment.⁷⁴

The relative benefit of sequential chemoradiation and concurrent chemoradiation was compared with radiation alone in a more recently reported STARTS (Comparison of Different Subsequent Treatments After Radical Surgery) trial patients with early-stage cervical cancer. Treatment with postoperative sequential chemotherapy followed by radiation was associated with a higher rate of DFS than treatment with adjuvant radiation therapy alone (3-year rate, 90.0% vs 82.0%; HR, 0.52) or concurrent chemoradiation (90.0% vs 85.0%; HR, 0.65; 95% CI, 0.44–0.96).¹³⁶ Although provocative, the trial included a subset who received neoadjuvant chemotherapy and included both intermediate- and high-risk patients together, which raises the question of for which patients sequential approaches should be considered. NRG 0724 recently completed accrual of high-risk patients treated with radical hysterectomy followed by chemoradiation with and without additional chemotherapy and may provide more insights on the impact of intensified chemotherapy in this population.¹⁵⁴

External-beam radiation for post-operative treatment typically includes the upper half of the vagina, the paravaginal tissues, and the pelvic and common iliac lymph nodes. The upper border of the field is typically at the top of the L5 vertebral body or may extend up to the bifurcation of the aorta. In cases in which treatment is primarily addressing paravaginal risk and nodal risk is low, such as a case with negative lymph nodes after a full dissection but indications to treat based on a cervical extension, a lower upper border may be considered. Small bowel fills the pelvis above the vaginal apex, so intensity-modulated radiation therapy (IMRT) can be used to shape the dose around the nodal target and avoid delivering prescription dose to the small bowel low in the pelvis (Fig. 48.6).

Definitive Radiotherapy for Stage IB1 and IB2 Disease

Radiotherapy alone also achieves excellent survival and pelvic disease control rates in patients with stage 1 and 2 cervical cancer (Fig. 48.7).¹⁵⁵ Eifel et al.¹⁵⁵ reported 5-year disease-specific survival and pelvic control rates of 90% and 98%, respectively, for 701 patients treated with radiation alone. Patients with stage IB3

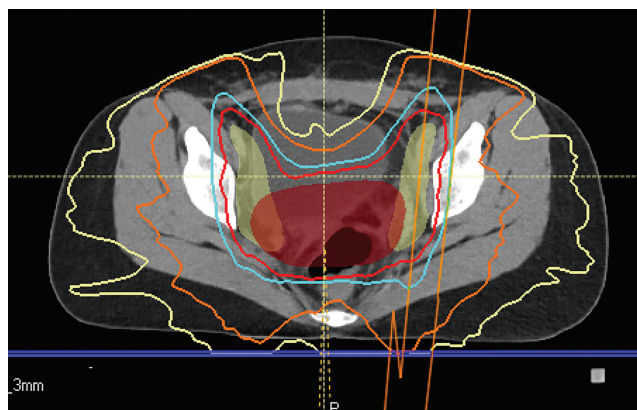


Figure 48.6 Typical intensity-modulated radiation therapy treatment plan for postoperative treatment of patients with cervical cancer. The nodal and vaginal target volumes are indicated in yellow and red, respectively. The prescription isodose line of 45 Gy is shown in red.

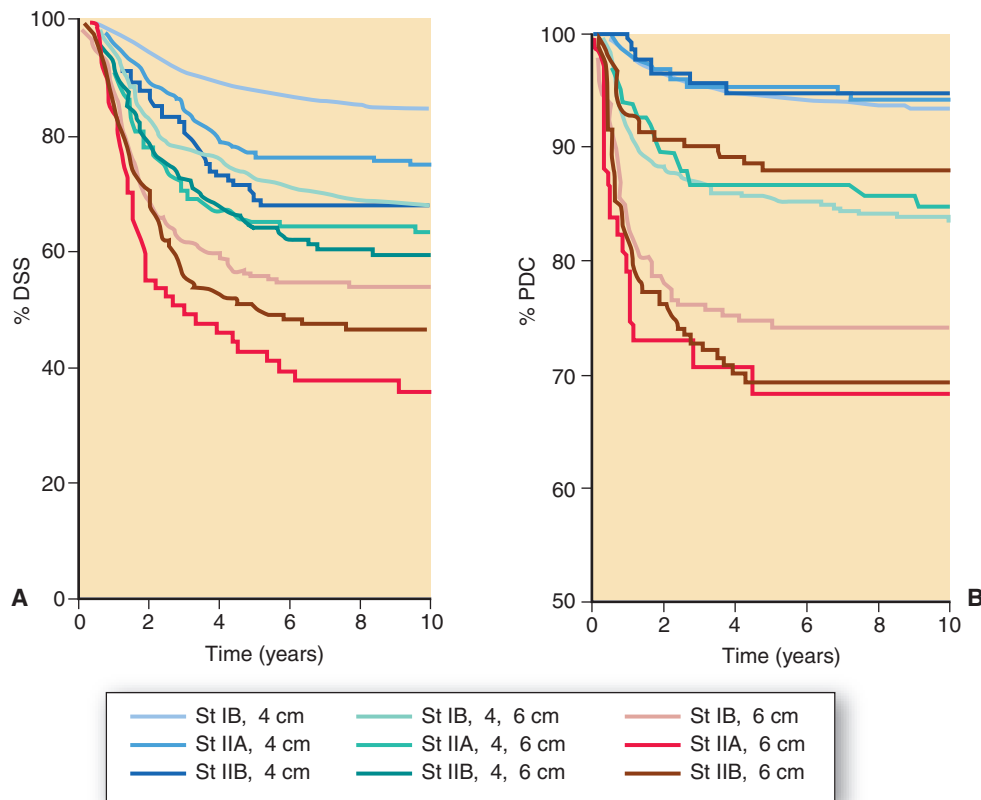


Figure 48.7 Disease-specific survival (DSS; **A**) and pelvic disease control (PDC; **B**) rates for 4,490 patients with stage I or II carcinomas of the cervix divided according to clinical tumor diameter and International Federation of Gynecology and Obstetrics (1988) stage.

and bulky stage IIA cancers are usually treated with concurrent cisplatin-based chemoradiation, which has been demonstrated in randomized trials to yield better outcomes than radiotherapy alone.¹⁵⁵

As with radical surgery, the goal of radical radiotherapy is to sterilize disease in the cervix, paracervical tissues, and regional lymph nodes in the pelvis. Patients are usually treated with a combination of external-beam irradiation to the pelvis and brachytherapy. Brachytherapy is used to increase the dose to the cervix. As discussed later in this chapter, most experts now recommend that the radiation dose be prescribed to a high-risk target volume encompassing the cervix as seen on MRI or CT rather than to a reference point (e.g., point A).¹⁵⁶ Patients with relatively small stage IB1 tumors may be treated to 80 to 90 Gy to this high-risk cervical target volume without exceeding doses to the bladder and rectum and thus limiting the risk of long-term toxicity.

Radiation Therapy Followed by Hysterectomy

Central tumor recurrence rates are relatively low, on the order of 10%, even for large tumors treated with radiation alone, suggesting that the margin for improvement with adjuvant hysterectomy is small.¹⁵⁵ In 2003, the GOG reported results of a prospective randomized trial of irradiation with or without adjuvant extrafascial hysterectomy in patients with stage IB tumors 4 cm or more in diameter; the study demonstrated no significant improvement in the survival rate among patients who had adjuvant hysterectomy (relative risk of death, 0.89; 95% CI, 0.65 to 1.21).¹⁵⁷ As a result, hysterectomy is not routinely recommended after chemoradiation, but it may still play a role in selected patients in whom uterine fibroids or other anatomic variations limit the dose of radiation deliverable with brachytherapy such as patients who have involvement of the uterine fundus with cancer. In these cases, extrafascial

(type I) hysterectomy is usually performed, in which the uterus, cervix, adjacent tissues, and a small cuff of the upper vagina in a plane outside the pubocervical fascia are removed. To minimize the risk of postoperative complications, the dose of brachytherapy must be decreased by about 25% if adjuvant hysterectomy is planned; radical hysterectomy is avoided after high-dose irradiation because of an increased risk of fistula and other complications.

Stage IB3, IIB, III, and IVA Disease

Radiation treatment in combination with chemotherapy is recommended for treatment of all locally advanced cervical cancers based on a series of randomized trials (see Table 48.3). Radiation treatment is delivered with an initial course of daily fractionated external-beam radiotherapy, targeting the uterus and upper vagina as well as at risk lymphatics. External-beam radiotherapy typically results in significant reduction in size of cervical cancers, which facilitates and improves dosimetry for subsequent brachytherapy. With appropriate chemoradiotherapy, even patients with locoregionally extensive cancers have a good chance of being cured of their disease.^{156,158}

In all cases, a course of external-beam irradiation (combined with concurrent chemotherapy) is used to deliver an initial homogeneous dose to the primary cervical tumor and to potential sites of regional spread; this initial treatment may also improve the efficacy of subsequent intracavitary brachytherapy by shrinking the bulky tumor and bringing it within the range of the high-dose portion of the brachytherapy dose distribution. Brachytherapy, which is also a critical component of definitive radiation therapy, exploits the inverse square law to deliver a high dose to the cervix and paracervical tissues while minimizing the dose to adjacent normal tissues.

Breaks during or between external-beam and intracavitary treatments should be discouraged, and every effort should be made to complete the entire radiation treatment in less than 7 to 8 weeks.

Several studies have suggested that treatment courses longer than 8 weeks are associated with decreased pelvic disease control and survival rates. In one study, survival decreased 0.6%/day, and pelvic control decreased 0.7%/day for each additional day of treatment beyond 55 days for all stages of disease.¹⁵⁹

A 2004 analysis of patterns of care for cervical cancer patients treated across the United States in radiation oncology facilities revealed that a concerning high percentage of patients were not completing treatment within this time frame. Approximately 40% of patients treated in nonacademic facilities failed to complete treatment within 10 weeks. Although treatment protraction also tended to be a problem in academic centers, the percentage of patients completing treatment in less than 10 weeks (83%) was significantly greater.¹⁶⁰ Additionally, brachytherapy is underused in patients with cervical cancer with concerning trends suggesting that rates of brachytherapy are declining in some locations,¹⁶¹ which is associated with lower rates of survival.¹⁶² These findings highlight the logistical and therapeutic challenges associated with delivery of multimodality cervical cancer treatment and suggest that optimal delivery of cervical cancer treatment requires extensive coordination between teams to ensure all essential components of treatment are delivered with minimal treatment delays.

External-Beam Radiotherapy Technique for Intact Cervical Cancer

External-beam radiation is typically delivered to 45 Gy in 25 fractions to the uterine target and the at-risk lymphatics. Grossly involved lymph nodes may be boosted with either an integrated or a sequential approach. With highly conformal boosts that includes minimal adjacent bowel, a final total dose of 60 to 66 Gy may be delivered. Treatment approaches include IMRT or conformal treatment plans. IMRT reduces the dose to normal tissues, including the small bowel which results in decrease in symptom burden, including a reduction in diarrhea and dysuria. It also permits dose escalation to grossly enlarged lymph nodes, which can reduce the overall treatment time. Examples of a IMRT plan for intact cervical cancer are shown in Figure 48.8. The increased conformality of the dose distribution around the nodal and vaginal target for a IMRT plan requires that internal organ motion is considered in defining contoured target volumes. The uterus can move several centimeters between treatment days based on changes in bladder or rectal filling. Broadly, two approaches are used to address this challenge when intact cervical cancer is treated with external-beam radiotherapy. One approach is to define an Internal Target Volume for the uterus, which accounts for all positions the uterus may take within the pelvis. Motion of the uterus caused by bladder filling can be assessed by imaging patients with the bladder full and empty at the time of simulation. Changes in rectal filling are more challenging to reproduce and may require a hypothetical ITV, created by assuming how much volume the rectum would occupy when empty of full and ensuring that the cervix would be included in the volume for both scenarios. The other approach is to choose a state of bladder filling at the time of simulation and perform daily image guidance before treatment to ensure that the uterus is within the target volume. This requires evaluation of images before treatment by someone with appropriate training in review of daily images.

Pelvic radiation can be delivered with a four-field conformal technique. These treatment fields typically include the whole low pelvis so that the uterus will be included, regardless of changes in bladder and rectal filling. This is typically accomplished with high-energy photons (15 to 18 mV) to spare superficial tissues. At these energies, the pelvis can be treated either with four fields (anterior, posterior, and lateral fields) or with anterior and posterior fields alone. When high-energy beams are not available, four fields are usually used because less penetrating 4- to 6-MV photons

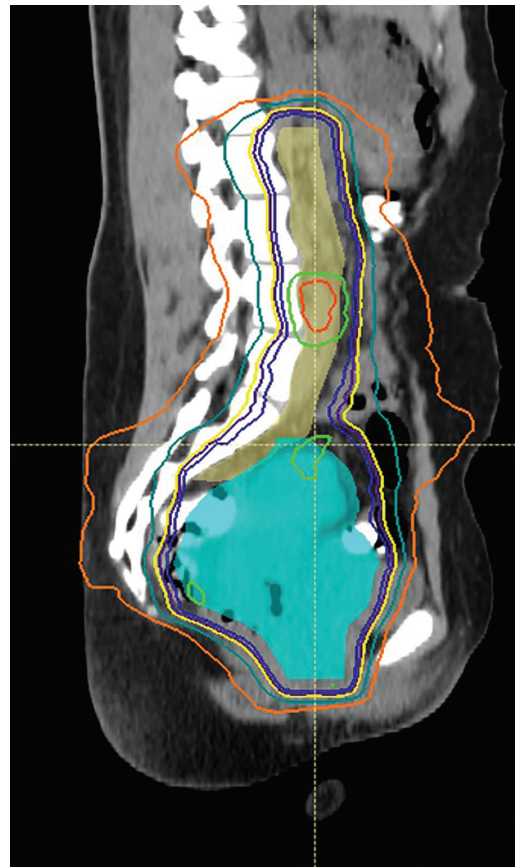


Figure 48.8 Sagittal intensity-modulated radiation therapy treatment plan for a patient with locally advanced cervical cancer and grossly involved pelvic and paraortic nodes. A central pelvic Internal Target Volume is shown in blue which encompasses the motion of the uterus with variation in bladder filling. The nodal clinical target volume (CTV) is shown in yellow. Forty-five Gy is prescribed (blue) to the uterine and nodal CTV with a boost to 50 Gy to the enlarged paraortic lymph node (red).

often deliver an unacceptably high dose to superficial tissues when only two fields are used.

Computed tomography simulation with or without contrast is performed to delineate the target volume, which includes the uterus, the vagina with margins on the lower extent of disease, and the at-risk lymphatics. High-quality imaging helps to delineate the target volume. MRI is particularly helpful in defining the extent of the gross tumor within the cervix and vagina as well as visualizing with more precision the site of the pathologically involved lymph nodes.¹⁶³ It can also help to determine if the ovaries are abnormal, which can be helpful in patients in whom the ovaries appear to be FDG avid on PET scan, which can be physiologic for premenopausal women or less likely represent metastasis to the ovaries. Additionally, MRI can provide the highest resolution imaging to define FDG-avid lymph sites in pelvis. Ultrasound gel in the vaginal can produce a bright hyperintense signal on T2-weighted MRI, which can help distinguish vaginal involvement from exophytic extension of tumor into the vaginal vault. MRI complements physical examination in this regard, which also critical for assessing vaginal extent of disease. PET can evaluate disease extent in the uterus but is also particularly helpful at identifying lymph node metastasis and distant spread. Identification of positive nodes relies on multiple lines of evidence; pathological enlargement, round without a fatty hilum, and a high-risk location are all signs that the lymph node is grossly involved and requires coverage with margin and subsequent boosting to bring the lymph node to a definitive dose. Dissemination to lymph nodes generally

follows a predictable pattern so that the location of a lymph node can be a factor in determining if it is pathologically involved.¹⁶⁵ The obturator nodes lying adjacent to the cervix are most commonly involved. Second and third echelon sites include the common iliac nodes and paraaortic nodes. Inguinal nodes are typically seen in patients with involvement of the distal vagina. Identifying suspicious nodes is critical for successful treatment as pathologically enlarged nodes, which are not boosted are very likely to persist and recur in the form of a pelvic side wall recurrence.

The inferior extent of disease can be determined by reviewing MRIs or by inserting radiopaque markers in the cervix or at the distal extent of vaginal disease. This is important to document at the beginning of treatment since extension of tumor into the vagina often resolves by the time of brachytherapy. In the case of a tumor that involves the vaginal wall mucosa, treatment with a cylinder may be necessary to ensure that there is sufficient dose to cover any residual microscopic disease. For patients with exophytic tumors that extend into the vagina from the cervix, cervical applicators such as ovoids or rings are sufficient to cover this disease extent, and a vaginal cylinder is not needed, reducing the risk of unnecessary vaginal stenosis.

Different strategies for defining the cervical cancer target volume are recommended by different groups but they also share the critical of ensuring that the at-risk tissues (cervix, uterus, upper vagina, parametria, and uterosacral ligaments when involved) are included in the target volume and that these targets will be covered in daily treatment. The EMBRACE (Image-guided intensity modulated External beam radiochemotherapy and MRI-based adaptive BRachytherapy in locally advanced CErvical Cancer) protocol recommends defining three target volumes for planning. The GTV-T is the gross target volume within the cervix. CTV-T HR includes GTV-T and any remaining cervix not infiltrated by tumor, and CTV-T LR includes the CTV-T HR in addition to complete parametria bilaterally, the entire uterus, and the uninvolved vagina with a 20-mm margin measured from the most inferior position of the initial HR CTV-T along the vaginal axis and finally an expansion on the CTV-T HR plus a margin of about 5 mm anterior and posterior toward the bladder and rectum. In case of involvement of the pelvic wall, sacrouterine ligaments, mesorectum, or other involved structures, a 20-mm margin around the initial HR CTV-T should be extended into these structures. Any pathological lymph nodes in the parametrium may be included. After CTV/ITV volumes are defined, an additional expansion to account for daily setup variation, a Planning Target Volume, of 5 to 10 mm is added depending on the accuracy of set up.

The prescription to the PTV is typically 45 Gy in 25 fractions, the same as the lymph node CTV so that contours can be simplified into a single CTV. External-beam doses of more than 40 to 45 Gy to the central pelvis tend to compromise the dose deliverable to paracentral tissues with brachytherapy and increase the risk of late complications.

Tumor response should be evaluated with periodic pelvic examinations. This is helpful for planning brachytherapy. Poor responses may impact decisions about applicators to use as well as evaluate regression of vaginal disease to determine if extension was exophytic from the cervix of extending along the vaginal mucosa. Treatment response is also a strong predictor of OS.¹⁶⁶ MRI performed at the end of external-beam radiotherapy and before brachytherapy can guide brachytherapy treatment and provide prognostic information.

Internal organ motion can be significant and should be considered in contouring, treatment planning, and decisions about daily image guidance. The uterus and vagina can move 3 to 4 cm with bladder and rectal filling, and even greater excursion is possible with anteversion or retroversion of the uterus, which can occur spontaneously. Broadly, two strategies can be useful to address this. Daily image guidance with CT may be used to deliver a IMRT plan based on a full or empty bladder scan. The daily CT can ensure that the uterus is within this target. Alternatively, an ITV volume encompassing the full potential excursion of the uterus can be used

in treatment planning in which case daily image guidance can be dictated by the PTV margin used. With daily kV images, a PTV margin can typically be limited to only 5 mm to 7 mm.

There is no evidence that IMRT can safely be used as an alternative to brachytherapy for routine treatment of patients with intact cervical cancer. Although IMRT achieves very conformal dose distributions, it cannot accurately reproduce the high-dose gradients produced with intracavitary brachytherapy. This dosimetric observation is supported by clinical data. Han et al.¹⁶² found that case-matched patients treated with brachytherapy had higher rates of DFS than those treated with external-beam techniques.

Role of Paraaortic Irradiation

In the past, the prognosis for patients with grossly involved paraaortic lymph nodes was poor. However, survival rates appear to have improved with the use of concurrent chemotherapy, modern diagnostic imaging methods, and highly conformal radiation techniques. With IMRT and PET-assisted target volume definition, Osborne et al.⁷³ reported a paraaortic disease control rate of 96% in patients who received radiation therapy for cervical cancers with grossly involved paraaortic nodes; the 5-year disease-specific survival rate for these patients was 47%. The survival of patients with paraaortic metastases has also been correlated with the bulk of central disease and the extent and size of involved lymph nodes. The duodenum should be contoured and avoided for patients with grossly involved paraaortic nodes. Limiting the volume receiving 55 Gy (V55 < 15 cc) has been reported to reduce toxicity.¹⁶⁶ The side effects of extended-field radiotherapy, particularly when combined with concurrent chemotherapy, can be substantial, and the management of such patients requires close multidisciplinary collaboration.

Radiation therapy fields typically extend superior to the highest grossly involved lymph node by approximately 6 cm to include next echelon lymph nodes, which may have microscopic disease involvement. Extended radiation treatment fields increase the bowel dose and subsequently the risk of short- and long-term toxicity, so the decision to extend the radiation field to cover elective nodes must be balanced against the risk of future nodal recurrence marginal to the radiation field.

Concurrent Chemoradiation

The publication of a series of prospective randomized trials has provided compelling evidence that the addition of concurrent cisplatin-containing chemotherapy to standard radiotherapy reduces the risk of disease recurrence by as much as 50%.^{74,158,167-170} The benefit of concurrent chemotherapy was subsequently confirmed in several metaanalyses that used individual patient data from published and unpublished randomized trials.⁷⁹ Additionally, a large Canadian population-based study that demonstrated a significant improvement in survival of women with cervical cancer after the 1999 adoption of chemoradiation as a standard did indicate that the benefits of chemoradiation can be generalized beyond the trial setting.¹⁶⁹ The trials varied in terms of stage of disease, dose of radiation, and schedule of cisplatin and radiation, although eligibility was in all cases limited to patients with disease confined to the pelvis. Although the improvement achieved with concurrent chemotherapy appears to be greater for patients with stages IB2 or II disease, metaanalyses indicate that combined treatment also benefits patients with stage III or IVA disease.⁷⁹ The benefit of concurrent chemotherapy also appears to be independent of histologic subtype or grade.⁷⁹ Although most trials focused on patients receiving definitive radiation therapy for locoregionally advanced disease, the SWOG trial published by Peters et al.⁷⁴ also demonstrated a benefit for patients who were found to have high-risk features (positive nodes or surgical margins or parametrial involvement) after radical hysterectomy for early-stage disease. However, for patients who undergo definitive radiation therapy for clinically

node-negative stage IB1 disease, the potential risks and benefits of concurrent chemotherapy must be carefully weighed, particularly if the patient has concurrent medical problems; such patients were not included in any of the sentinel trials, in part because the margin for improvement over radiation therapy alone is very small for such patients.¹⁵⁵

Most of the chemoradiation trials had cisplatin-based regimens in the experimental arms, using either cisplatin alone (weekly at a dose of 40 mg/m²) or a combination of cisplatin (50 to 75 mg/m²) and 5-FU given every 3 weeks. Although both approaches appeared to be of benefit, the weekly cisplatin used in most GOG studies has emerged as the most frequently recommended standard, primarily because it is associated with less severe acute hematologic and gastrointestinal (GI) side effects than combined cisplatin and 5-FU. Other chemotherapy approaches have been tested and, in some cases, found to be of benefit. Several early trials demonstrated that combinations of mitomycin C and 5-FU with radiation improved outcomes over treatment with radiation alone.⁷⁹ However, the value of 5-FU as a single agent given concurrent with radiation has not been clearly established for patients with cervical cancer. A GOG trial comparing weekly cisplatin versus concurrent continuous infusion 5-FU was discontinued before achieving the planned accrual because futility analysis indicated that 5-FU was not going to be superior to cisplatin.¹⁷¹ Weekly cisplatin has never been directly compared with mitomycin and 5-FU or with the combination of every-3-week cisplatin (75 mg/m²) or 5-FU that was used in Radiation Therapy Oncology Group (RTOG) trial 90-01 or SWOG 87-97, a phase 3 randomized trial (NCT01101451) investigating the role of chemoradiation for stage I to IIA patients with intermediate-risk factors after treatment with radical hysterectomy.

Only one large randomized trial has compared multiagent concurrent chemoradiation with weekly cisplatin.¹⁷² In a large multinational trial, 500 women with stage IIB to stage IVA cervical cancer, were randomly assigned to the experimental treatment arm, which included cisplatin and gemcitabine with concurrent external-beam radiation followed by brachytherapy and adjuvant chemotherapy with gemcitabine and cisplatin. Women in the standard treatment group received cisplatin and concurrent external-beam radiation therapy followed by brachytherapy alone. The authors reported encouraging results, with a 3-year progression-free survival (PFS) of 74% for the combination chemotherapy regimen versus 65% with cisplatin alone ($P = 0.03$). However, in part because the trial had incomplete follow-up, particularly of late side effects, and because the combined regimen had more severe acute side effects than weekly cisplatin, this approach has not gained wide acceptance. The trial also left unanswered whether any improvement was due to the addition of concurrent gemcitabine or to the addition of postradiation adjuvant chemotherapy.¹⁷³

The OUTBACK trial was an international randomized phase III trial of the Gynecologic Cancer InterGroup. Eligible women had locally advanced cervical cancer (FIGO 2008 stage IB1 and node-positive IB2, II, IIIB, or IVA disease) that was suitable for primary treatment with chemoradiation with curative intent. Women were randomly assigned to either standard cisplatin-based chemoradiation (control) or standard cisplatin-based chemoradiation followed by adjuvant chemotherapy (experimental) with four cycles of carboplatin and paclitaxel after stratification for nodal status, participating site, FIGO stage, age, and planned extended-field radiotherapy. A total of 919 women were eligible and included in the primary analysis. OS at 5 years was similar in those assigned experiment versus control arms (72% vs 71%; $P = 0.91$; HR, 0.91, 95% CI, 0.70 to 1.18). Grade 3 to 5 adverse events within 1 year of randomization occurred in 81% in the experimental arm versus 62% in the control group.¹⁷⁴

Because the goal of adjuvant treatment is primarily to reduce the incidence of distant disease, several groups have tried to develop models that could predict a high risk of distant recurrence in patients treated with chemoradiation for locally advanced

cervical cancer. In a Korean Gynecology Oncology Group study,¹⁷⁵ a four-parameter model incorporating regional nodal findings on FDG PET, nonsquamous cell histology, and pretreatment serum SCC antigen levels was strongly predictive of distant recurrence, with a rate of approximately 60% for their highest risk group versus less than 25% for their low- and intermediate-risk groups.

Taken together, the randomized trials provide strong evidence that the addition of concurrent chemotherapy to pelvic radiotherapy benefits selected patients with locoregionally advanced cervical cancer. A metaanalysis of randomized trials confirmed the benefit of concurrent chemoradiation, showing an advantage for both cisplatin- and non-cisplatin-containing regimens.⁷⁹ However, most trials have explicitly excluded patients who had evidence of paraaortic lymph node metastases, poor performance status, or impaired renal function.

Brachytherapy

Brachytherapy was first used to treat patients with cervical cancers in the early 20th century and continues to play a central role in their curative management. The goal of brachytherapy is to deliver a high dose to disease in the cervix and paracervical tissues while preserving function of adjacent critical structures.

The uterine cavity provides an ideal receptacle for radioactive sources, which are positioned using specially designed applicator systems that capitalize on the distinctive anatomy of the distal female genital tract and the physical advantages of the inverse square law. Brachytherapy applicators consist of an angled or curved intra-uterine tandem with some form of intravaginal applicator; vaginal applicators used in various clinical settings include several versions of the Fletcher-Suit afterloading colpostats, vaginal rings, vaginal cylinders, and others. Vaginal packing or balloons are used to hold the applicator in place and to maximize the distance between the sources and the bladder and rectum. Intraprocedural image guidance is helpful in enduring applicators that are placed adjacent to the cervix. Ultrasound can be used to guide the placement of the applicator into the endometrial canal. This can be accomplished by placing fiducial markers and obtaining an orthogonal radiogram. Alternatively, CT can be used, which provides information about the position of the applicator within the uterus and the position of normal tissues relative to the applicators. MRI provides the highest resolution and can be used to determine the location of the cervical tumor relative to applicators. Regardless of which imaging modality is used, intraprocedural imaging is essential to make sure that the position of applicators is optimized; Optimization with treatment planning cannot overcome applicators that are not well positioned. Applicators should lie in proximity to tumor (ovoids and rings should lie flush against the cervix). The tandem should lie inside the uterine canal, extending to the top of the fundus or at least a couple of centimeters cranial to the cervical tumor. Any packing placed should lie between the applicators and bladder and rectum and not between the vaginal applicators and the cervix. If intraprocedural CT and MRI cannot be obtained, they can be obtained outside for the procedural space to help with treatment planning. MRI serves as the gold standard based on the highest soft tissue resolution and ability to delineate the target.

Treatments may be delivered over several days at a continuous low-dose rate (LDR), with frequent pulses (pulsed-dose rate [PDR]) or in several fractions of radiation delivered at a high-dose rate (HDR). HDR or PDR techniques that use remote afterloading units. These approaches use a single source of iridium-192 that travels or steps through the positions along the length of the tandem, ovoid, or needles within the applicators. The amount of time the source dwells within each of these positions can be determined as part of the process of optimizing a treatment plan. The activity of sources used for HDR is approximately 10 Ci; the activity of sources used for PDR is approximately 0.5 to 1 Ci. The applicator systems and rules of optimal brachytherapy placement are similar for LDR,

HDR, and PDR treatments. The importance of radiation dose rate and fraction size is discussed in more detail later in this section.

Radiation dose delivered by intracavitary brachytherapy declines rapidly as an inverse function from the distance from the intrauterine and intravaginal sources squared. These characteristic permits delivery of a relatively high dose to tissues close to the applicator with much lower doses delivered just a few cm away. Treatment plans are generated by focusing on specific points or volumes that are a few centimeters away from the applicator. It is important to remember that immediately adjacent to the applicators, doses will be much higher than a reference dose 1 or 2 cm or away; dose distribution is very heterogeneous with brachytherapy. The classic target reference point, point A, is defined as a point 2 cm lateral to external cervical os and 2 cm above the tops of the intravaginal applicators (Fig. 48.9). Point A was thought to reference the crossing of the ureter and the uterine artery, but the size of the tumor and cervix largely dictate the relationship between these structures and the applicators. In many patients, point A is the center of the tumor with the ureter and uterine artery pushed lateral, but for others with small tumors and cervixes, these structures may lie medial to point A. Switching to a dose prescription based on the individual points' anatomy has now become the standard, using the image-guided brachytherapy paradigm.

Image-Based Brachytherapy Treatment Planning

Computed tomography and MRI can be used to verify appropriate placement of the applicator and evaluate the dose to the target and adjacent critical structures. CT has the advantage of being more widely available than MRI and does not require the use of special MRI-compatible applicators. However, MRI provides much better visualization of soft tissues and more accurate appreciation of the

extent of tumor within the cervix and paracervical tissues^{177,178}. Incorporation of image guidance appears to improve outcomes for patients with cervical cancer. Higher rates of local control rates were seen in patients treated with a volume based target compared with a point-based prescription (92% vs 86%; $P = 0.01$)¹⁷⁸ in a meta-analysis, including patients with stage I to IVA cervical cancer treated with concomitant chemoradiotherapy and brachytherapy.

The Groupe Européen de Curiethérapie–European Society of Therapeutic Radiation Oncology (GEC-ESTRO) consortium has developed a series of guidelines on the optimal approach for imaging, delineating target and avoidance volumes, and treatment planning using MRI.^{8,17} According to their guidelines, T2 multiplanar images should be obtained after applicator placement to allow delineation of the GTV that is typically hyperintense on T2 images.¹⁶³ The high-risk clinical target volume (HR-CTV) encompasses the entire cervix and gross residual macroscopic disease at the time of brachytherapy. An intermediate-risk clinical target volume (IR-CTV) includes the initial extent of disease plus a 1-cm margin on the HR-CTV. Using these methods, Dimonopolous et al.¹⁷⁹ reported a strong correlation between the dose to the HR-CTV and local control, with a 96% control rate when the equivalent dose at 2 Gy per fraction (EQD2) to 90% of the target volume (D90) was 87 Gy or greater. GEC-ESTRO also recommends delivering a total dose equivalent of 60 Gy to the IR-CTV D90. However, in some cases, compromise may be necessary to avoid exceeding normal tissue tolerance limits. For each implant, tomographic images should be used to delineate and estimate the doses delivered to critical structures, including the bladder, rectum, and sigmoid. Current recommendations suggest limiting the minimum dose to the maximally irradiated 2 cc (D2cc) of the bladder to less than 80 Gy and the minimum dose to the rectum and sigmoid to less than 70 Gy,¹⁵⁶ although these dose relationships have been most convincingly demonstrated for

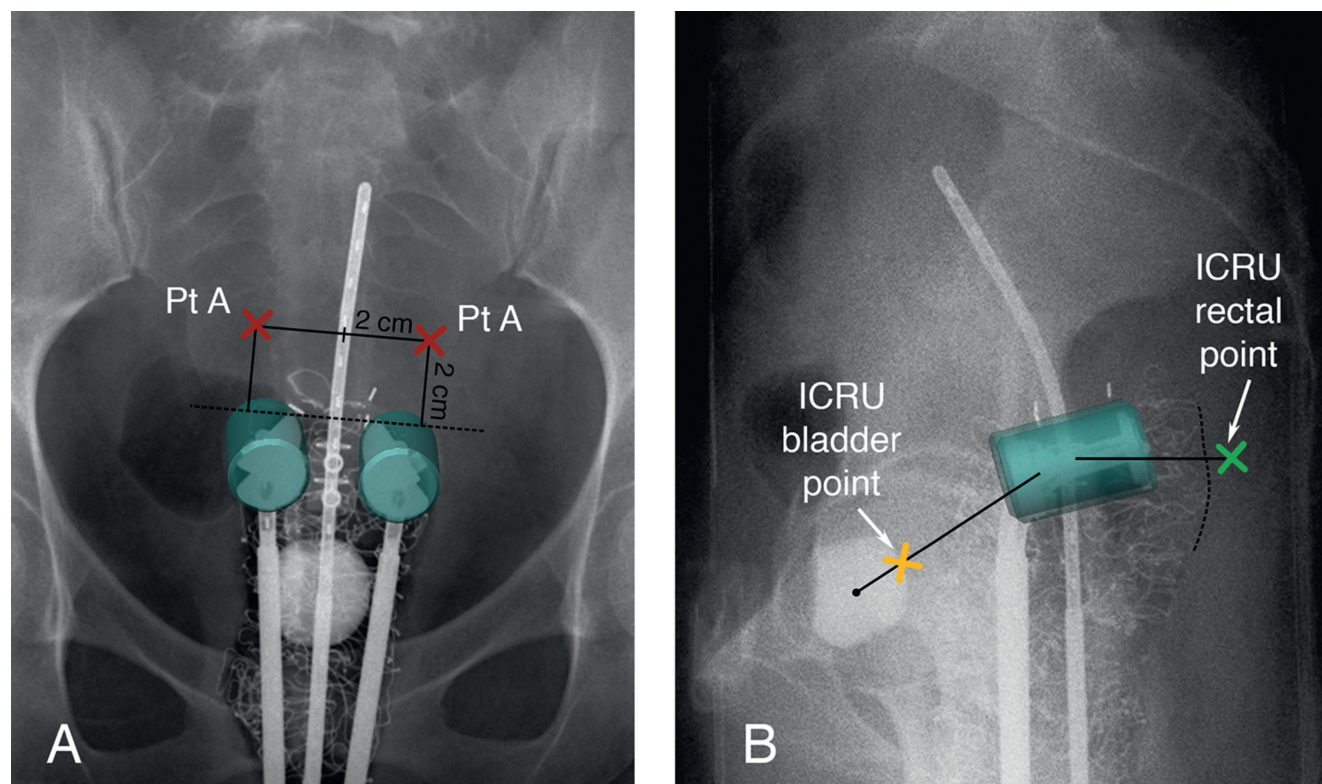


Figure 48.9 Posteroanterior and lateral views of a Fletcher-Williamson applicator system in a patient with invasive carcinoma of the cervix. Traditional reference points used to report dose to point A (Pt A), bladder, and rectum are shown. The teal overlay depicts the contours of the ovoids. Radiopaque threads in the gauze used to pack the vagina reveal the contours of the vagina after applicator placement. ICRU, International Commission on Radiation Units and Measurements. (With permission from Eifel PJ, Klopp AH. *Gynecologic Radiation Oncology: A Practical Guide*. Wolters Kluwer; 2017.)

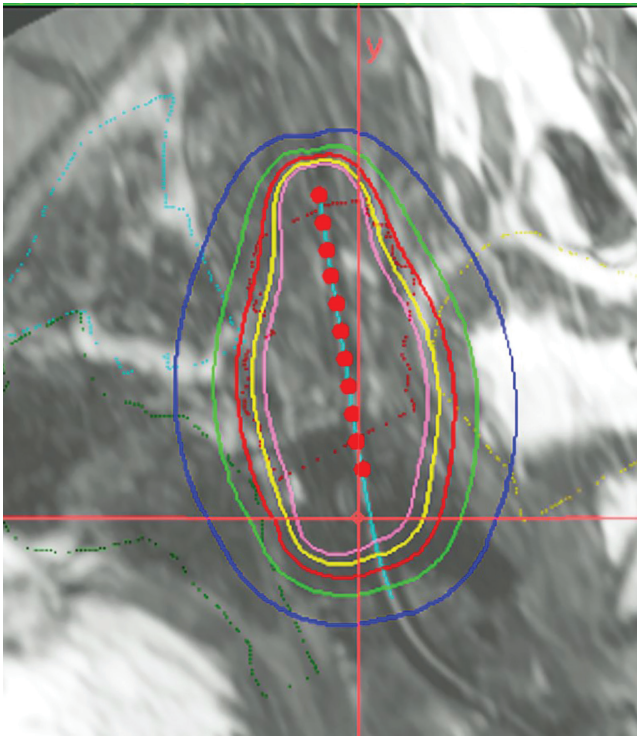


Figure 48.10 Magnetic resonance imaging (MRI)-optimized brachytherapy treatment plan. Dwell positions within intrauterine tandem are shown as filled red circles. The prescription dose is shown as a red line, encompassing the high-risk clinical target volume, which includes gross residual disease on MRI as well as the entire uterine cervix.

the rectum. Results of the prospective multicenter EMBRACE (International Study on MRI-Guided Brachytherapy in Locally Advanced Cervical Cancer) trial suggest that an even lower rectal D2cc of 65 Gy or less may yield further reduction in the risk of radiation proctitis.¹⁸⁰ An example of an MRI-based plan, optimized to cover the HR-CTV while minimizing dose to normal tissues, is shown in Figure 48.10.

In 2012, Charra-Brunaud et al.¹⁸¹ reported the results of a prospective, nonrandomized, multicenter comparison of 705 patients treated for cervical cancer using two-dimensional or three-dimensional (3D) image-guided brachytherapy techniques. Treatment in the 3D group was performed according to GEC-ESTRO guidelines that were adapted for CT. Patients were assigned to one of three groups: patients receiving brachytherapy alone followed by surgery; patients receiving external-beam radiotherapy, brachytherapy, and surgery; or patients with more advanced cervical cancers treated with external-beam radiotherapy and brachytherapy. In all three groups, the local relapse-free survival rate was higher in the 3D group, and the rates of grade 3 and 4 complications were lower. Because treatment approach was determined by the practices of the respective centers, other factors may have differed between the centers. In addition, the D90 doses delivered to the HR-CTV in 117 patients who received definitive radiotherapy planned using 3D techniques (median, 73.1 Gy) were considerably lower than the dose of 87 Gy recommended by GEC-ESTRO¹⁷⁹; this may have contributed to a relatively high central recurrence rate of 23% at 2 years. However, these results do support the safety and effectiveness of image-guided brachytherapy.

Brachytherapy Dose Rate and Treatment Schedule

Cervical cancer brachytherapy can be delivered with an LDR, PDR, or HDR approach. LDR brachytherapy was the standard for

almost 100 years; with this approach, treatment is delivered continuously, usually over 24 hours to 72 hours using cesium-137 sources that are manually inserted in the hollow applicators after they have been optimally positioned in the uterus and vagina. Typically, treatment is given after an initial course of pelvic radiotherapy and is usually administered in two sessions separated by about 2 weeks. PDR brachytherapy follows a similar treatment schedule as LDR; however, the treatment is delivered using a single remotely controlled source of iridium (called a “stepping source”) that travels through the applicators delivering the prescribed treatment in hourly pulses over several days.¹⁸² With HDR brachytherapy, a very high-activity iridium stepping source (~10 Ci) is used to deliver a high dose of radiation over several minutes. HDR brachytherapy is usually delivered in four to seven treatments spaced over 2 weeks to 5 weeks. Again, treatment is almost always integrated with a course of pelvic external-beam irradiation.

Traditional LDR brachytherapy has been performed with sources that deliver the dose of radiation at a rate of approximately 0.4 Gy to 0.6 Gy per hour to point A. These LDRs permit repair of sublethal cellular injury with preferential sparing of normal tissues. In a randomized trial, Haie-Meder et al.¹³² reported a significant increase in treatment-related complications when the dose rate was doubled from 0.4 to 0.8 Gy per hour, indicating that the total dose must be reduced and that the therapeutic ratio of treatment may be compromised with higher dose rates in some cases. Differences in the magnitude of the dose-rate effect between tumor and normal tissues may partly reflect differences in the half-times for repair of sublethal radiation damage.⁷⁴

There have been few direct comparisons between LDR and HDR treatment of patients with cervical cancer. Early prospective trials were criticized for methodologic flaws and used methods that have limited relevance today. However, there have now been many observational studies published that document the effectiveness of concurrent chemoradiation given using a combination of external-beam radiotherapy and image-guided HDR intracavitary brachytherapy. Survival rates appear to be similar to those reported using LDR brachytherapy. As with LDR treatment, successful HDR brachytherapy depends on optimized applicator position, balanced use of external-beam therapy and brachytherapy, compact overall treatment duration, and delivery of an adequate dose to tumor with respect for normal tissue tolerance limits.

Intestinal brachytherapy involves the placement of needles directly into the tissue, as contrasted with intracavitary implants, in which applicators are placed within cavities such as the uterus and vagina. Sources can step through these interstitial needles in the same way that they do for intracavitary uterine and vaginal applicators. This additional flexibility in location of radiation sources possible to cover more complex and irregular dose targets. Interstitial needles are used within perineal templates that contain numerous paralleled needles placed around an intravaginal obturator and are advanced from the perineum. A disadvantage of these applicators is the fact that they do not take advantage of the natural cavity within the cervix, which is so ideal for shaping dose distributions around the cervix. The advantages of both intracavitary and interstitial applicators can be captured with hybrid applicators that have needles that traverse past the vaginal applicators lying adjacent to the cervix and into the uterine cervix and parametrial tissues.¹⁸² Well-positioned needles make it possible to escalate dose or reduce normal tissue doses with less risk of placing needles into critical adjacent organs.¹⁸³ An example of this approach comparing dose distribution from a standard intracavitary applicator with a hybrid applicator is shown in Figure 48.11. An alternative approach to place needles within the cervix to shape dose around asymmetric volumes is the use of freehand interstitial needles can also be placed within the cervix. This approach may require more experience and necessitates high-quality intraprocedural imaging to be sure the needles are well positioned.

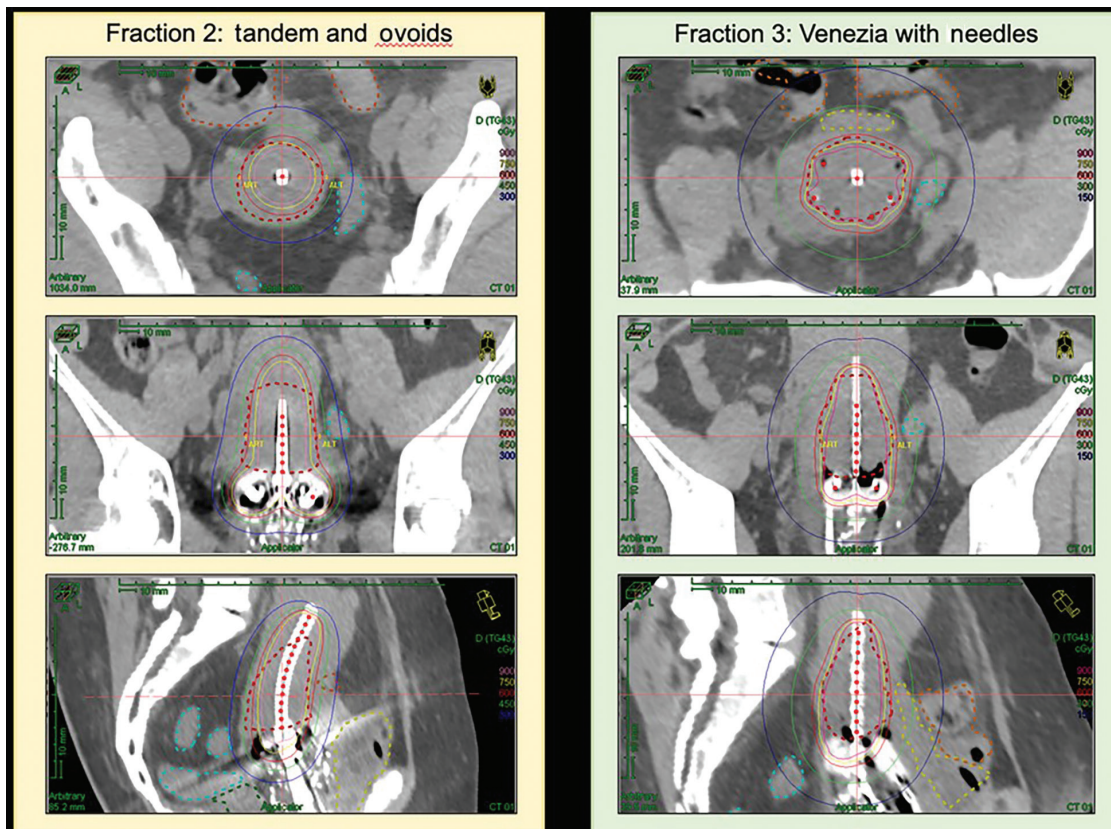


Figure 48.11 Axial image of a computed tomography plan from a patient receiving high-dose-rate brachytherapy with an intracavitary applicator (**left**) for her second fraction and with a hybrid applicator (**right**) for her third fraction. Placement of the interstitial needles allowed more lateral distribution of the prescription dose without increasing normal tissue doses.

Complications of Radiotherapy

During pelvic radiotherapy, most patients have mild fatigue and mild to moderate diarrhea that usually is controllable with antidiarrheal medications; some patients have mild bladder irritation, which may be symptomatic of a urinary tract infection.¹⁸⁴ When extended fields are treated, patients may have nausea, gastric irritation, and depression of peripheral blood cell counts. Hematologic and GI complications are significantly increased in patients receiving concurrent chemotherapy.¹⁵⁸ Unless the ovaries have been transposed, all premenopausal patients who receive pelvic radiotherapy experience ovarian failure by the completion of treatment.

Perioperative complications of intracavitary brachytherapy include uterine perforation, fever, and the usual risks of anesthesia. Thromboembolisms are rare. In a review of 4,043 patients who had 7,662 intracavitary applications for cervical cancer, Jhingran and Eifel reported 11 patients (0.3%) with thromboembolisms, 4 of whom—all with pulmonary embolisms—died.¹⁸⁶ All 4 fatal pulmonary embolisms were in patients with advanced pelvic wall disease. However, appropriate prophylaxis should be used to prevent deep vein thrombosis in all patients who will have prolonged immobilization.

Estimates of the risk of late complications of radical radiotherapy vary according to the grading system, duration of follow-up, method of calculation, treatment method, and prevalence of risk factors in the study population. However, most reports quote an overall risk of major complications (requiring transfusion, hospitalization, or surgical intervention) of 5% to 15%.^{155,158,184,185} In a study of 1,784 patients with stage IB disease, Eifel et al.¹⁸⁴ reported an overall actuarial risk of major complications of 7.7% at 5 years. Although the actuarial risk was greatest during the first 3 years of follow-up, there was a continuing risk to surviving patients of

approximately 0.3% per year, resulting in an overall actuarial risk of 14% at 20 years. Complication rates may be higher in patients with very locally advanced disease in part because of tissue destruction caused by infiltrative tumor.

During the first 3 years after treatment, rectal complications are most common and include bleeding, stricture, ulceration, and fistula. In the study by Eifel et al.,¹⁶⁰ the risk of major rectosigmoid complications was 2.3% at 5 years. Major GI complications were rare 3 years or more after treatment, but a constant low risk of urinary tract complications persisted for many years (Fig. 48.12). The actuarial risk of developing a fistula of any type was 1.7% at 5 years. Complication rates were calculated actuarially, and patients who died without experiencing a major complication were censored at the time of death.

Small bowel obstruction is an infrequent complication of standard radiotherapy for patients without special risk factors. The risk of small bowel obstruction is significantly increased in patients who have undergone an open transperitoneal lymph node dissection.⁷⁰ Other factors that can increase the risk of small bowel complications in patients treated for cervical cancer include a history of pelvic inflammatory disease or peritonitis, thin body habitus, heavy smoking, and the use of high doses or large volumes for external-beam irradiation, particularly with low-energy treatment beams and large daily fraction sizes.¹⁸⁶

Numerous psychological and physical factors can influence sexual function after pelvic radiation therapy.¹⁸⁷ Most patients who receive definitive radiation therapy for cervical cancer have some agglutination and telangiectasia of the apical vagina. More significant vaginal shortening can occur, particularly in older, postmenopausal women and those with extensive tumors treated with a high dose of radiation. Hypoestrogenism can enhance vaginal atrophy and dryness, contributing to dyspareunia. Intravaginal or systemic

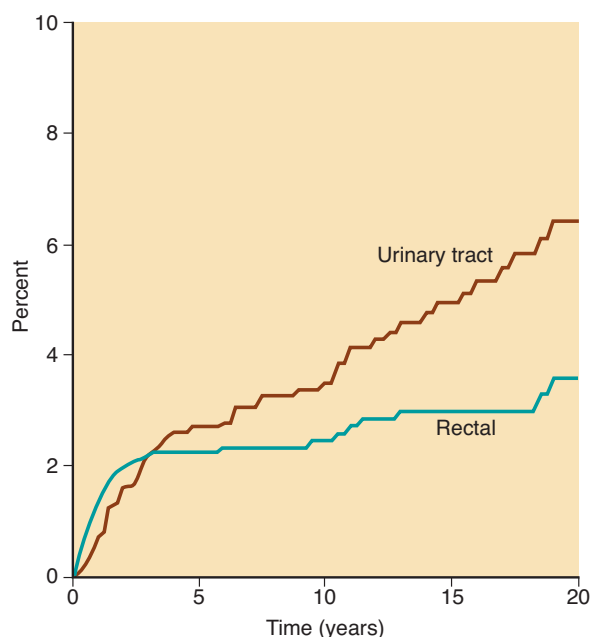


Figure 48.12 Rates of major rectal and urinary tract complications in 1,784 patients with stage IB carcinomas of the cervix treated with radiotherapy. (From Eifel PJ, Levenback C, Wharton JT, et al. Time course and incidence of late complications in patients treated with radiation therapy for FIGO stage IB carcinoma of the uterine cervix. *Int J Radiat Oncol Biol Phys* 1995;32(5):1289–1300, with permission.)

estrogen may reduce these symptoms. Vaginal dilation may be helpful at preventing sexual dysfunction.^{188,189}

Radiotherapy in Patients with Metastatic Disease

Localized radiotherapy can provide effective relief of pain caused by metastases in the bone, brain, lymph nodes, or other sites. Patients with progressive paraaortic nodes may have severe pain, particularly when involving the spine or psoas muscle and may have marked pain relief with radiation treatment. A rapid course of pelvic radiotherapy can also provide excellent relief of pelvic pain and bleeding for patients who present with incurable disseminated disease. Hypofractionated courses of treatment, such as a single conformal treatment of 8 or 10 Gy, can rapidly relieve vaginal bleeding and pain.¹⁹⁰

In some cases, patients with metastatic disease may have disease restricted to a small number of sites that can be feasibly treated with radiation therapy to curative doses, such as lymph nodes in the mediastinum, supraclavicular area, or limited-volume lung metastasis. Ablative treatment to multiple sites of disease has been shown to lead to long-term survival in a subset of patients but has not yet been tested in a randomized fashion for patients with cervical cancer.¹⁹¹

Recurrence After Definitive Irradiation

Some patients who have an isolated central recurrence after radiotherapy can be cured with surgical treatment. Because the extent of disease may be difficult to evaluate and the risk of serious urinary tract complications of pelvic surgery is high after high-dose radiotherapy, surgical salvage treatment usually requires a pelvic exenteration, most often an anterior or total exenteration. Less extensive operations, such as radical hysterectomy, are reserved for selected patients with small tumors confined to the cervix or lesions that do not encroach on the rectum.¹⁹²

Careful patient selection and counseling are of paramount importance when considering someone for pelvic exenteration. Regardless of the indication, patients undergoing a pelvic exenteration must be in otherwise good medical health to be able to tolerate a long surgical procedure with extensive fluid shifts and prolonged hospital stay. Part of the evaluation process includes comprehensive assessment to exclude unresectable or metastatic disease. MRI and/or PET/CT evaluation should be performed to exclude substantial pelvic sidewall involvement or extrapelvic metastasis.¹⁹³ Tumor involvement of the pelvic sidewall is a contraindication for exenteration but may be difficult to assess if there is extensive radiation fibrosis. Patients should be counseled about changes in body image and function. Specifically, patients should have an understanding of anatomic changes involving creation of colostomy and urinary conduit and need to be accepting of major changes in body image even in the setting of reconstructive surgery. Patients require significant family support, intact mental capacity, and access to continued and long-term medical care. Long-term survival is directly correlated with complete tumor resection, so establishing resectability is a key aspect of preoperative planning. The 5-year survival rate for those who undergo pelvic exenteration is 20% to 60%.^{194,195}

The exploratory operation begins with a thorough inspection of the abdomen for evidence of intraperitoneal spread, lymph node metastases, or pelvic sidewall infiltration. Despite careful preoperative evaluation, about 30% of operations are aborted intraoperatively. Frozen section biopsies are done of suspicious areas. If the biopsy findings are negative, the surgeon proceeds to remove the bladder, rectum, vagina, uterus, ovaries, fallopian tubes, and all other supporting tissues in the true pelvis. A urinary conduit, a transverse or sigmoid colostomy, and a neovagina are created.^{195,197} Postoperative recuperation may take several months. The surgical mortality rate is less than 10%, with most postoperative complications and deaths related to sepsis, pulmonary thromboembolism, and intestinal complications such as small bowel obstruction and fistula formation. The rate of GI complications may be reduced by using unirradiated segments of bowel and by closing pelvic floor defects with omentum, rectosigmoid colon, or myocutaneous flaps. Advances in low colorectal anastomosis and techniques for creating continent urinary reservoirs have improved the quality of life for selected patients. Most investigators report that the most common problems for survivors relate to their adjustment to urostomy or colostomy. Women with vaginal reconstruction tend to report fewer sexual problems and better quality of life than those without reconstruction.^{196,197} All of these findings indicate the importance of adequate counseling after the exenterative surgery.

For patients who have unresectable recurrent disease after definitive irradiation, treatment options are limited. Most patients who have unresectable pelvic recurrences after radiotherapy are treated with chemotherapy alone; response rates and prognosis are generally poor.

Systemic Treatments

Patients who present or relapse with disease in distant organs or who have extensive recurrence in previously irradiated sites are usually incurable. The care of these patients must emphasize palliation of symptoms with use of appropriate pain medications and localized radiotherapy.

Single-Agent Chemotherapy

Cisplatin has been studied in a variety of doses and schedules in the management of patients with recurrent or metastatic cervical cancer and is considered the most active agent against this malignancy.¹⁹⁸ Although a number of other agents (e.g., ifosfamide, carboplatin, irinotecan, and paclitaxel) have exhibited a modest level of biologic activity in cervical cancer (producing response rates of 10%–15%),¹⁹⁹ the clinical utility of these drugs in patients who

have not responded to cisplatin or who have experienced recurrence or progression after chemoradiation is uncertain. Furthermore, it is well recognized that the objective rate of response to chemotherapy is lower in previously irradiated areas (e.g., pelvis) than in nonirradiated sites (e.g., lung).²⁰⁰

Combination Chemotherapy and Targeted Therapy

Most reports of combination chemotherapy for carcinoma of the cervix have described small, uncontrolled phase II trials of drug combinations. The results of two phase III randomized trials, published in 2004 and 2005, provided the first solid evidence that combination chemotherapy can improve both PFS (cisplatin plus paclitaxel vs single-agent cisplatin,²⁰¹ cisplatin plus topotecan vs single-agent cisplatin) and OS (cisplatin plus topotecan versus single-agent cisplatin) when it is administered for recurrent or metastatic carcinoma of the cervix.²⁰² However, the phase III GOG-204 trial comparing combinations of cisplatin with topotecan, paclitaxel, gemcitabine, or vinorelbine revealed no significant differences in outcome between patients treated with the four cisplatin-based regimens.²⁰³

Carboplatin is less toxic than cisplatin in terms of nephrotoxicity, neurotoxicity, and emetogenicity, so the combination of carboplatin and paclitaxel has been evaluated in this setting. A multicenter phase II study showed that the carboplatin and paclitaxel regimen was feasible and had similar efficacy as other cisplatin-based doublets for the treatment of metastatic or recurrent cervical cancer.²⁰⁴ Furthermore, in the Japan Clinical Oncology Group (JCOG-0505) randomized phase III trial of paclitaxel plus carboplatin versus paclitaxel plus cisplatin in patients with stage IVB, persistent, or recurrent cervical cancer, the carboplatin doublet was noninferior to cisplatin doublet in terms of OS; however, among patients who had not received prior cisplatin (including as chemoradiation), the cisplatin doublet was associated with increased OS compared with the carboplatin doublet, suggesting that cisplatin remains the key drug for patients who have not previously received platinum agents.²⁰⁵

Tumor angiogenesis plays a significant role in the progression of cervical cancer and has been associated with a poor prognosis. Bevacizumab prevents tumor angiogenesis by blocking vascular endothelial growth factor and was shown to be active in a phase II study (GOG-227C) in recurrent cervical cancer.²⁰⁶ The GOG-240 study explored the addition of bevacizumab to chemotherapy in a randomized phase III trial with a 2 × 2 factorial design in which OS was the primary endpoint. Patients with primary stage IVB or recurrent or persistent disease were randomized to paclitaxel–cisplatin or paclitaxel–topotecan, both with or without bevacizumab. Two main conclusions were obtained from this study: first, the median OS is significantly prolonged by the addition of bevacizumab (16.8 months vs 13.3 months; HR, 0.765; 95% CI, 0.62 to 0.95; *P* = 0.006) and second, nonplatinum doublet is not superior to cisplatin–paclitaxel even in the population previously treated with cisplatin. Patients treated with bevacizumab had a higher risk of grade 2 or greater hypertension (25% vs 1.8%), grade 3 or greater venous thromboembolic events (8.2% vs 1.8%), and grade 2 or greater fistula (8.6% vs 1%), and these side effects must be carefully monitored during treatment.²⁰⁷

Immunotherapy

The presence of HPV-specific CD4⁺ helper and CD8⁺ cytotoxic T cells has been reported in cervical tumors, highlighting the inherent immunogenicity of these tumors. Co-gain or co-amplification of both programmed cell death protein ligand 1 (PD-L1) (CD274) and programmed cell death protein ligand 2 (PD-L2) (PDC-DLG2) by fluorescence in situ hybridization has been identified in 67% of cervical cancers and has been associated with higher PD-L1 protein expression by immunohistochemistry.²⁰⁸ As a

strategy to reverse tumor-mediated immune resistance, agents targeting immune checkpoints, most commonly PD-1, PD-L1, and cytotoxic T-lymphocyte-associated protein 4, have been explored across a number of disease settings.

In the KEYNOTE-158 phase II trial, in the subset of 98 patients with previously treated advanced cervical cancer, the objective response rate (ORR) was 12.2% (95% CI, 6.5% to 20.4%), with three complete and nine partial responses. All 12 responses were in patients with PD-L1–positive tumors, for an ORR of 14.6% (95% CI, 7.8% to 24.2%) in the PD-L1–positive population; 14.3% (95% CI, 7.4% to 24.1%) of these responses were in those who had received one or more lines of chemotherapy for recurrent or metastatic disease.²⁰⁹ The results were comparable to the outcomes of the phase I/II CHECKMATE-358 trial, a multicohort study of nivolumab with or without ipilimumab for virus-associated cancers. In the monotherapy arm of this trial, an overall response rate of 26.3% and 20% was observed in the 19 treated patients with cervical cancer and the 5 treated patients with vulvar or vaginal cancer, respectively.²¹⁰

In a double-blind, phase 3 randomized controlled trial, women with patients with persistent, recurrent, or metastatic cervical cancer were randomly assigned in a 1:1 ratio to receive pembrolizumab (200 mg) or placebo every 3 weeks for up to 35 cycles plus platinum-based chemotherapy and, per investigator discretion, bevacizumab. The dual primary endpoints were PFS and OS. In 548 patients with a PD-L1 combined positive score of 1 or more, median PFS times were 10.4 months in the pembrolizumab group and 8.2 months in the placebo group (HR, 0.62; 95% CI, 0.50 to 0.77; *P* < 0.001). In 617 patients in the intention-to-treat population, PFS times were 10.4 months and 8.2 months, respectively (HR, 0.65; 95% CI, 0.53 to 0.79; *P* < 0.001). The most common grade 3 to 5 adverse events were anemia (30.3% in the pembrolizumab group and 26.9% in the placebo group) and neutropenia (12.4% and 9.7%, respectively).²¹¹ In October 2021, the FDA approved pembrolizumab in combination with chemotherapy with or without bevacizumab for patients with persistent, recurrent or metastatic cervical cancer whose tumors express PD-L1 (combined positive score ≥1), as determined by an FDA-approved test.

Adoptive T-Cell Therapy

Approaches based on infusion of ex vivo–expanded tumor-infiltrating lymphocytes (TILs) and T cells genetically modified to express HPV-targeted T-cell receptors (TCRs) have been explored in several studies.²¹² In a large phase II trial of TIL therapy conducted by the National Institutes of Health, responses were observed in 28% among 18 patients with cervical cancer.²¹³ An interim analysis of another phase II trial of TIL therapy in 27 patients with advanced cervical cancer demonstrated a response rate of 44%, including 1 complete response, 9 partial responses, and 2 unconfirmed partial responses. The disease control rate was 89%. At a median follow-up period of 3.5 months, 11 of the 12 responding patients maintained their responses.²¹⁴ Based on these results, in May 2019, the FDA granted LN-145 breakthrough designation for cervical cancer. Although TILs have higher concentration of specific T cells compared with peripheral T cells, the hostile tumor microenvironment attenuates the long-term survival of functional T cells because TILs are sensitive to anergy, exhaustion, and apoptosis.

The infection with high-risk human papillomavirus (HR-HPV) is a noted driver for the development of nearly all cervical cancers. E6 and E7 oncoproteins are highly expressed by HPV-positive cervical cancer cells, becoming attractive therapeutic targets for engineered T cells. A phase I/II study of human leukocyte antigen (HLA) A2–restricted E6 T-TCR-modified T cells for HPV-associated cancers has been reported. Among 6 patients with cervical cancer, 2 displayed stable disease. In the phase I portion, no severe adverse effects were observed.²¹⁵ Chimeric antigen receptor T cell (CAR-T) therapy, in contrast to TCR T cell therapy, can directly